

Sentiment and Thematic Analysis of "My One NZ" Mobile Application User Reviews

A Business Data Analytics Project

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1. Introduction & Problem Identification

Context: The Competitive Telecom Landscape

One NZ (rebranded from Vodafone) operates in a highly competitive New Zealand telecom market. Mobile apps are now critical customer touchpoints, with 90% of users expecting a smooth experience. Poor app performance directly impacts customer lifetime value and competitive advantage.

The Business Challenge

The My One NZ app faces a customer satisfaction crisis due to persistent issues: authentication failures, instability, and billing inconsistencies. This results in low ratings and potential churn.

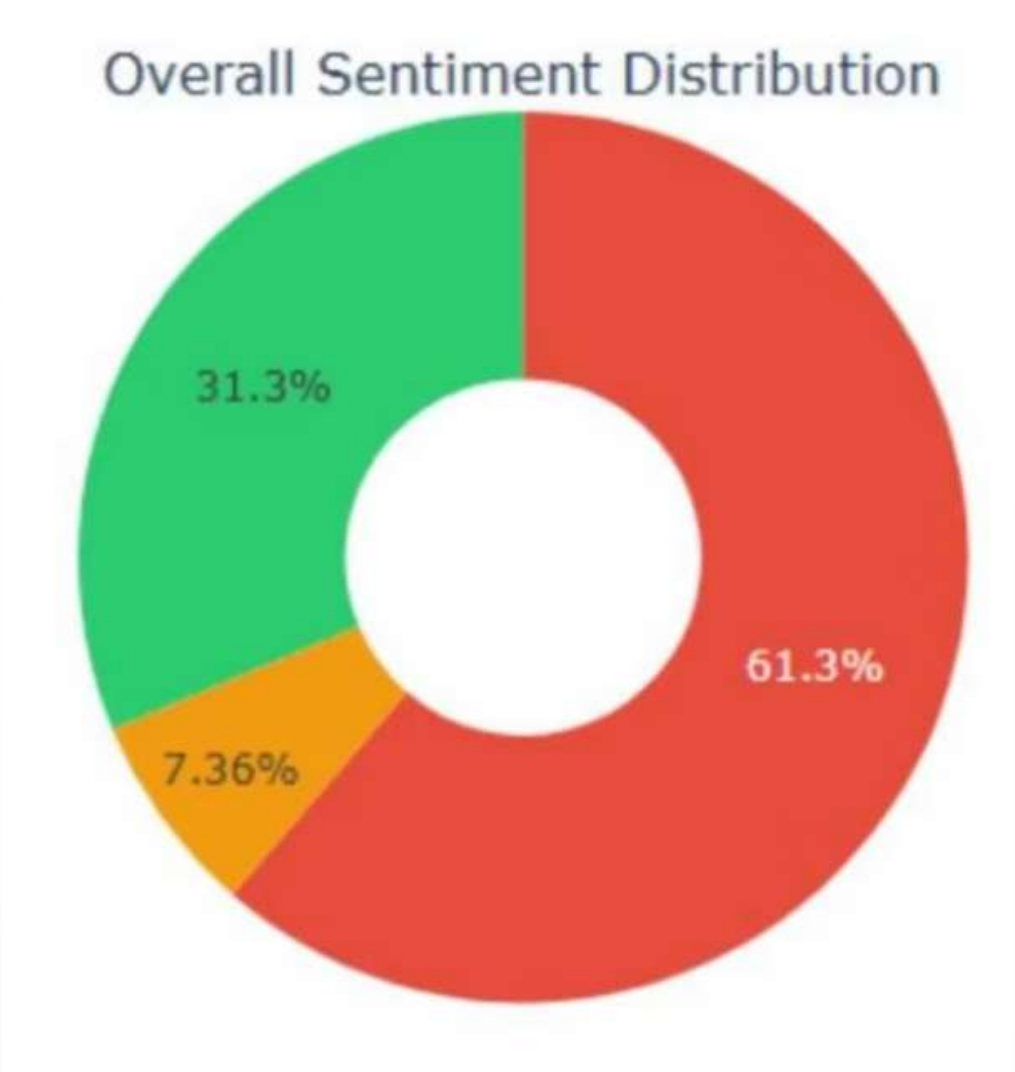
61%

Negative Sentiment

\$9-15M

Annual Financial Exposure

Overall Sentiment Distribution



The majority of reviews (61.3%) express negative sentiment, indicating widespread user dissatisfaction.

Key Questions Addressed



What specific issues drive negative sentiment and their frequency?



Which application versions demonstrate highest dissatisfaction?



How effective are current customer response strategies?



Can we predict negative reviews for early intervention?



What actions will maximize satisfaction within resource constraints?



2. Literature Review: Analytical Foundations

Sentiment Analysis in BI

Leveraging customer feedback for competitive advantage and product prioritization. Ensemble methods often outperform individual techniques.

ML for Text Classification

Progression from rule-based (VADER) to deep learning models (BERT, RoBERTa) that capture bidirectional context for sentiment.

NLP Techniques

Computational methods for text interpretation, including VSM, TF-IDF, and contextual embeddings from neural language models.

App Analytics

Reviews are vital unstructured data. Functional failures (35% of negative reviews) and feature-level sentiment provide actionable insights.

3. Methodology: Model Architecture & Data Pipeline

Data Collection & Preprocessing 9,386 Google Play Store reviews (Dec 2011–Sep 2025) were collected. Raw data was processed through a four-stage NLP pipeline to optimize for different analyses.

The study uses quantitative secondary data analysis, combining descriptive statistics, supervised classification, and unsupervised topic discovery for robust validation.

- **Data Collection**
9,386 public Google Play Store reviews (Dec 2011–Sep 2025).
- **Preprocessing & NLP**
Normalization, tokenization, stopwords removal, and lemmatization to structure raw text.
- **Sentiment Analysis**
Comparative analysis using VADER, TextBlob, and RoBERTa deep learning model.
- **Supervised ML**
TF-IDF vectorization used to train five algorithms (Logistic Regression, Random Forest, SVM, Naive Bayes, and Gradient Boosting).
- **Topic Modeling**
Latent Dirichlet Allocation (LDA) and Non-Negative Matrix Factorization (NMF) to discover latent thematic structures (e.g., update, data, billing).
Keyword Analysis to identify the issues related to telecom domain

4. Tools Used and Data Visualization

Programming & Analytics Tools

- **Python 3.10, Pycharm, Jupyter**
Core environment for development and interactive analysis.
- **Pandas & NumPy**
Data manipulation and numerical operations.
- **NLTK, TextBlob, Hugging Face**
Natural Language Processing and deep learning models.
- **Scikit-learn & Gensim**
Machine learning classification and topic modeling (LDA/NMF).

Visualization Techniques

A variety of visuals were used to convey complex data clearly:

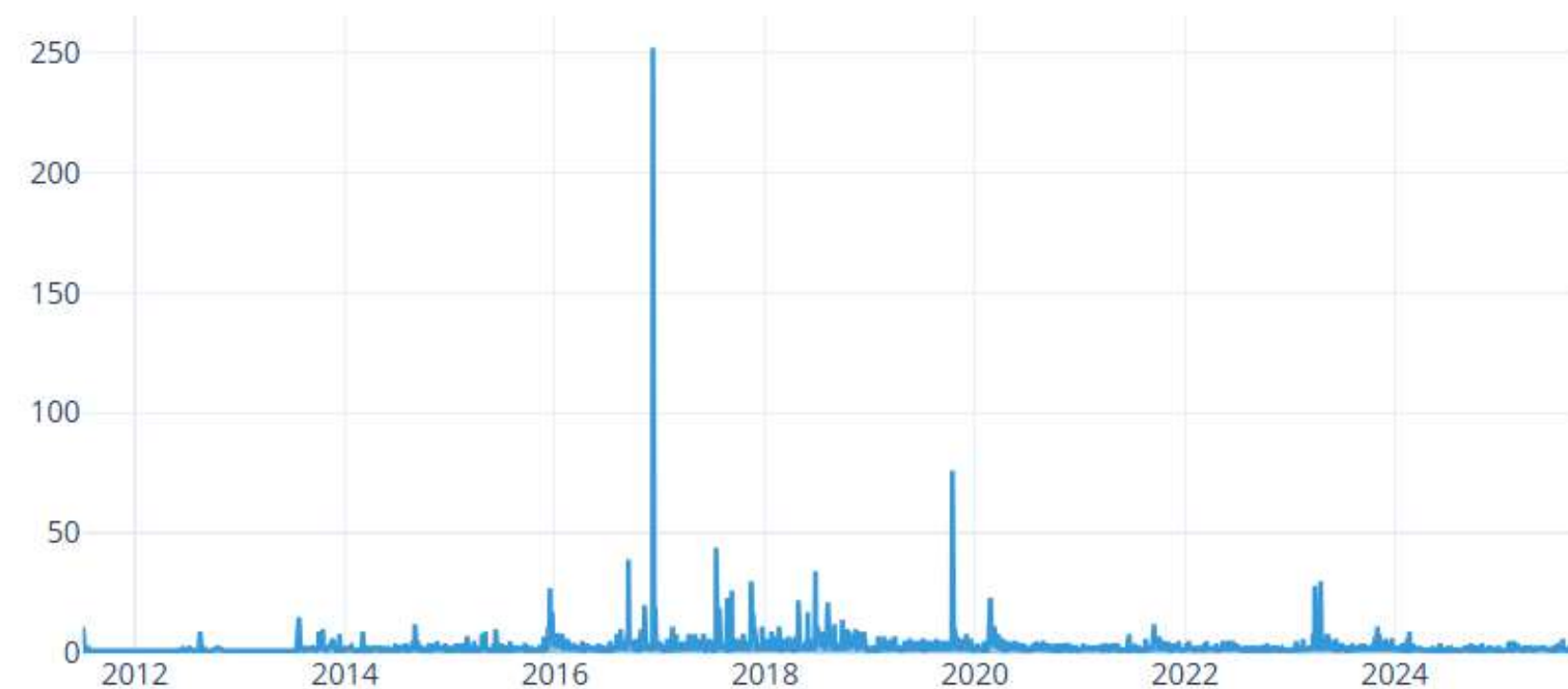
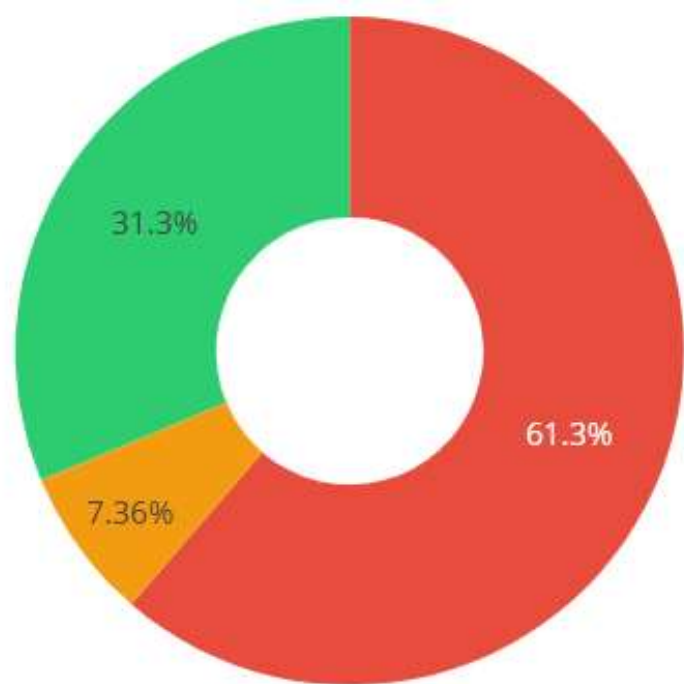
- Word Clouds (TF-IDF weighted frequency)
- Time Series Charts (with moving averages)
- Stacked Area Charts (sentiment breakdown)
- Violin Plots and Annotated Heatmaps (Confusion Matrices)



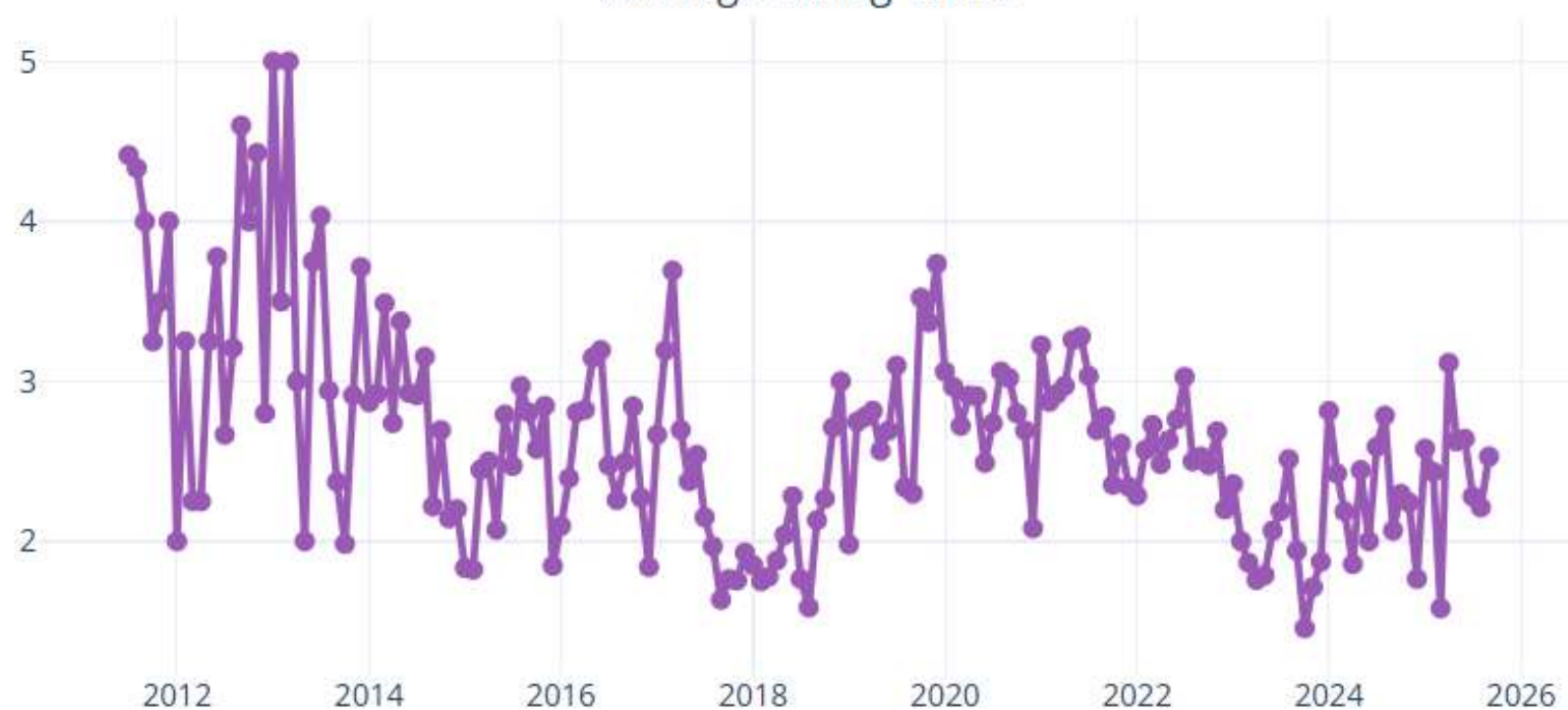
5. Critical Findings - Sentiment Crisis

			
61.3%	2.73	79.1%	6000+
Negative Sentiment	Mean Rating	RoBERTa Accuracy	Low Response Rate
5,756 reviews show crisis-level negative sentiment.	Significantly low compared to competitor Spark NZ (4.4).	Deep learning model achieved the highest sentiment classification accuracy.	A high volume of reviews received no official response.

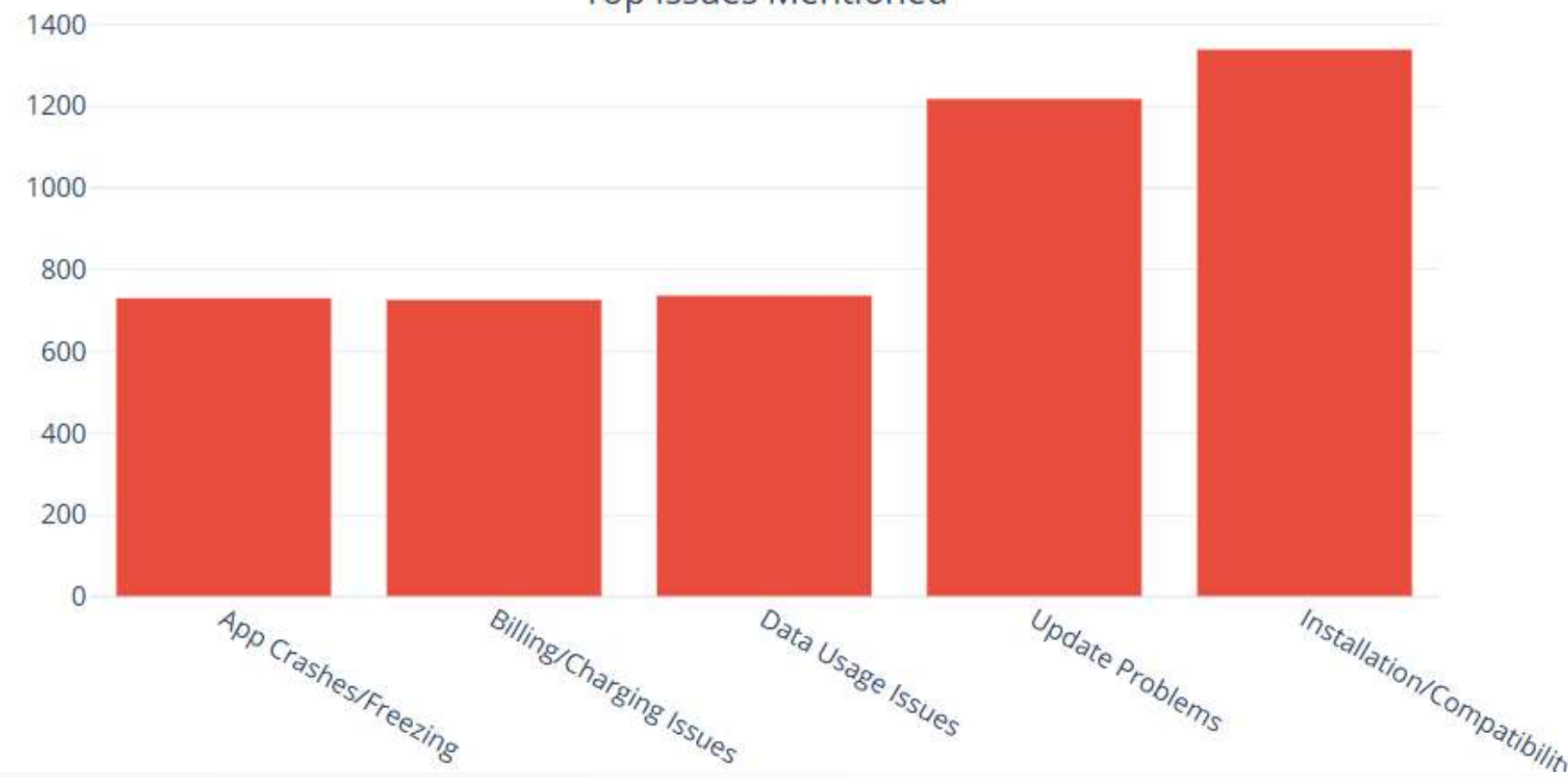
Version 3.x generated the most negative reviews, while Version 7.x currently holds a low 2.45 rating.



Average Rating Trend



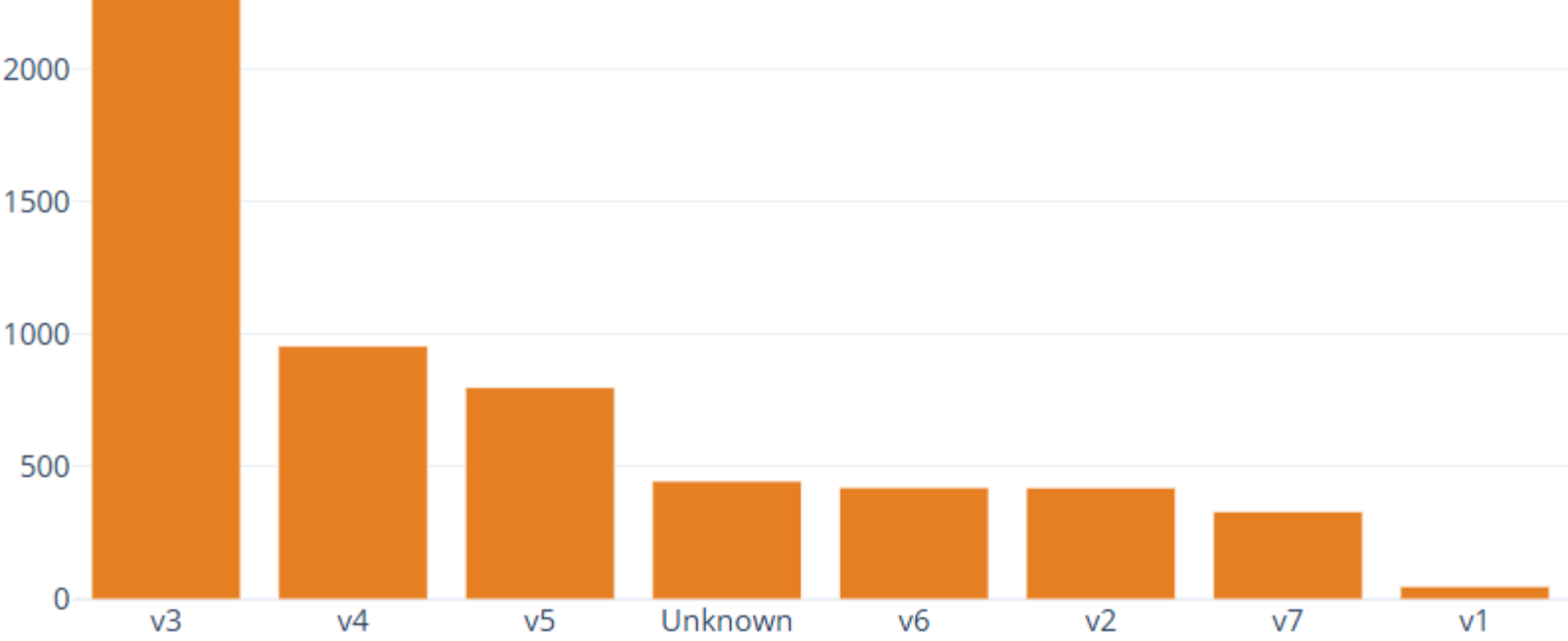
Top Issues Mentioned

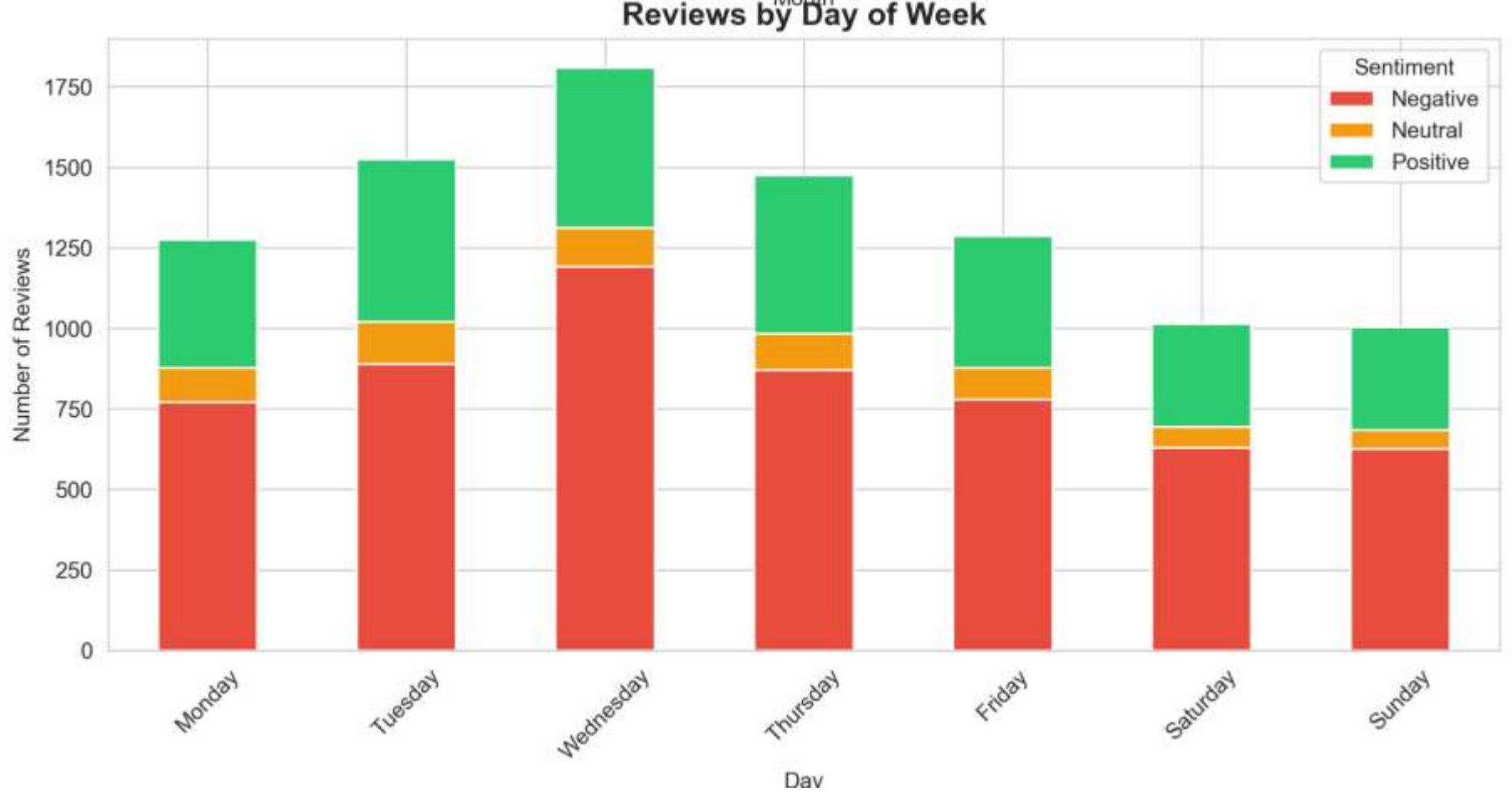
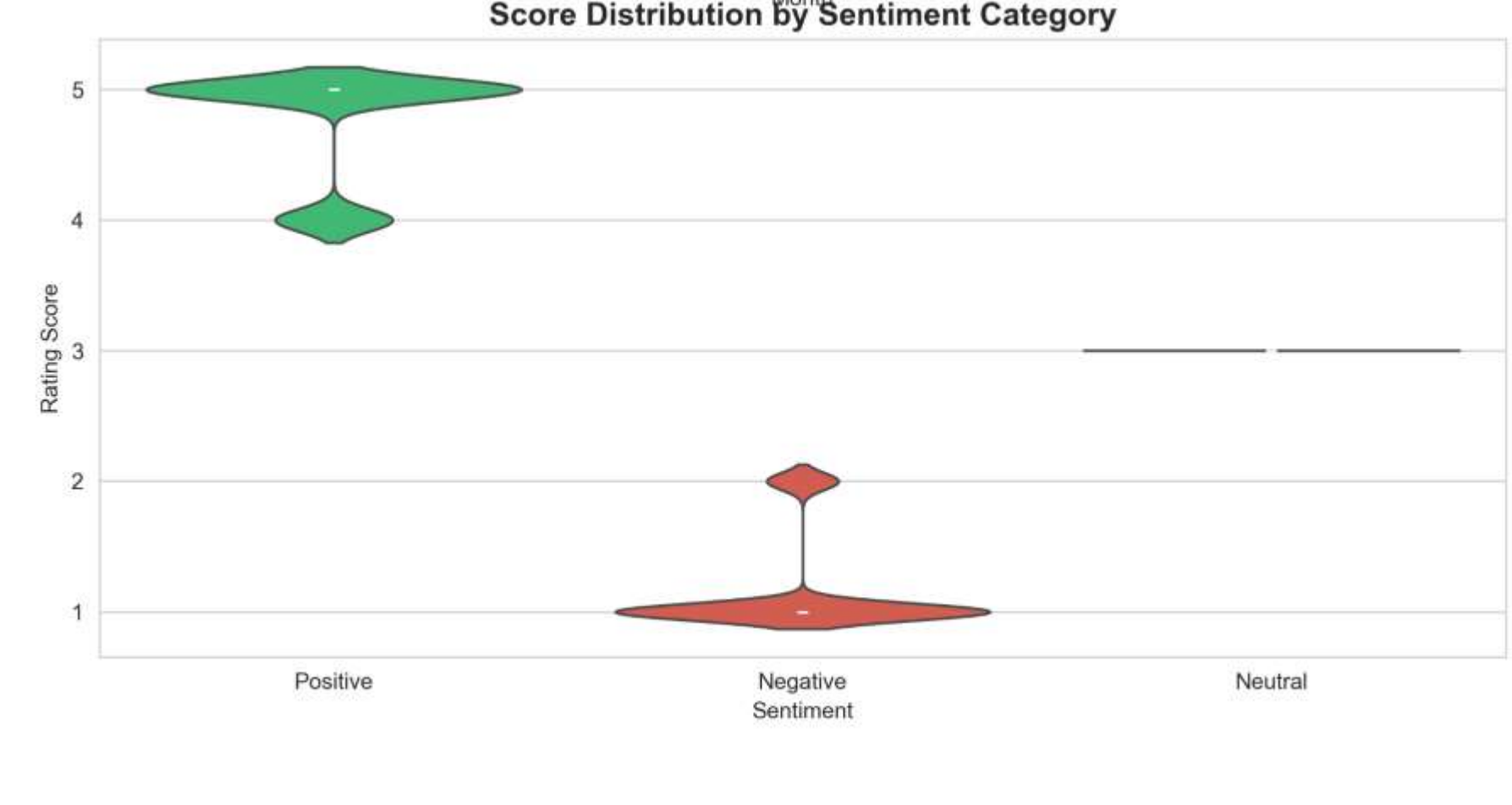
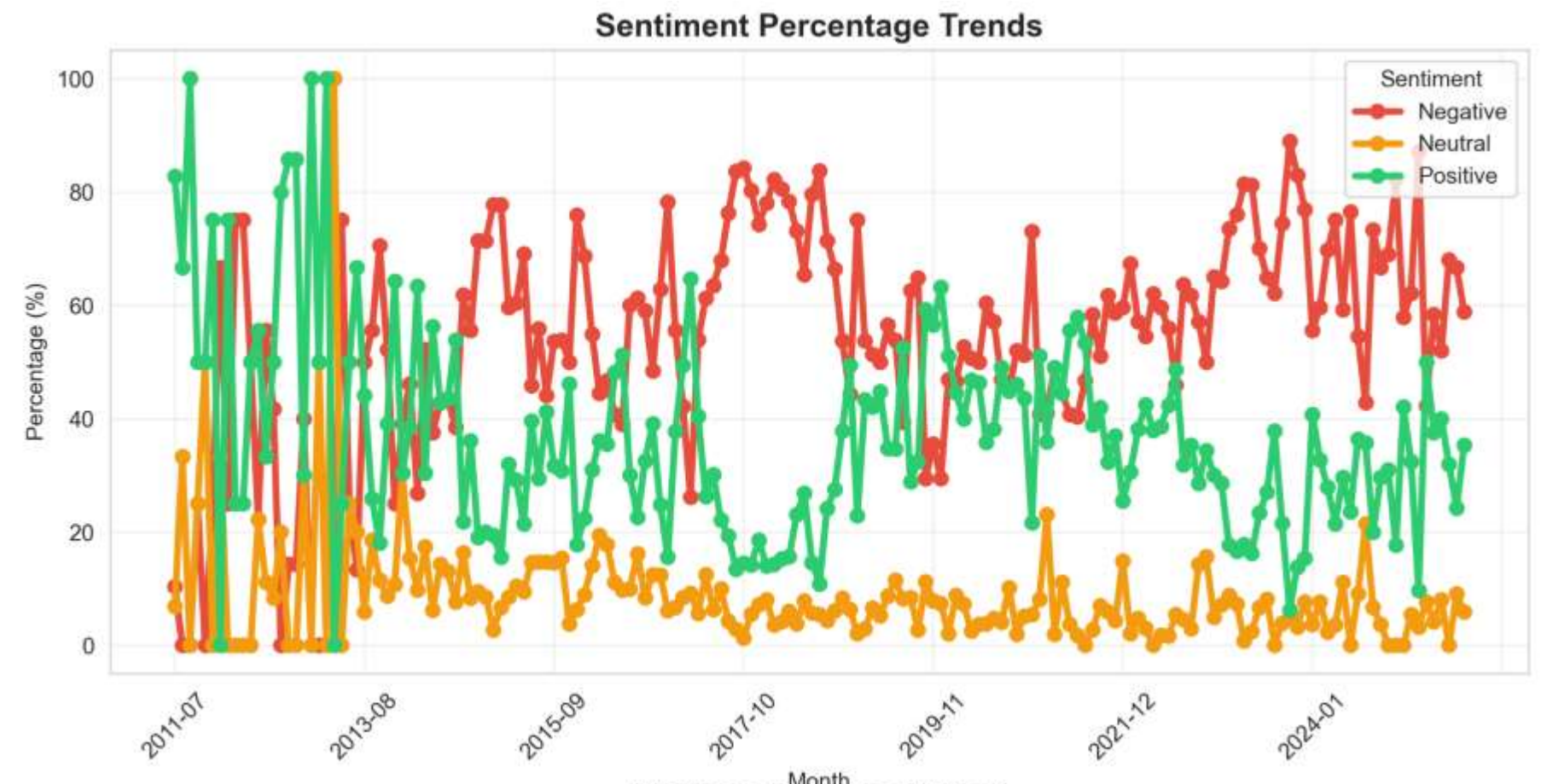
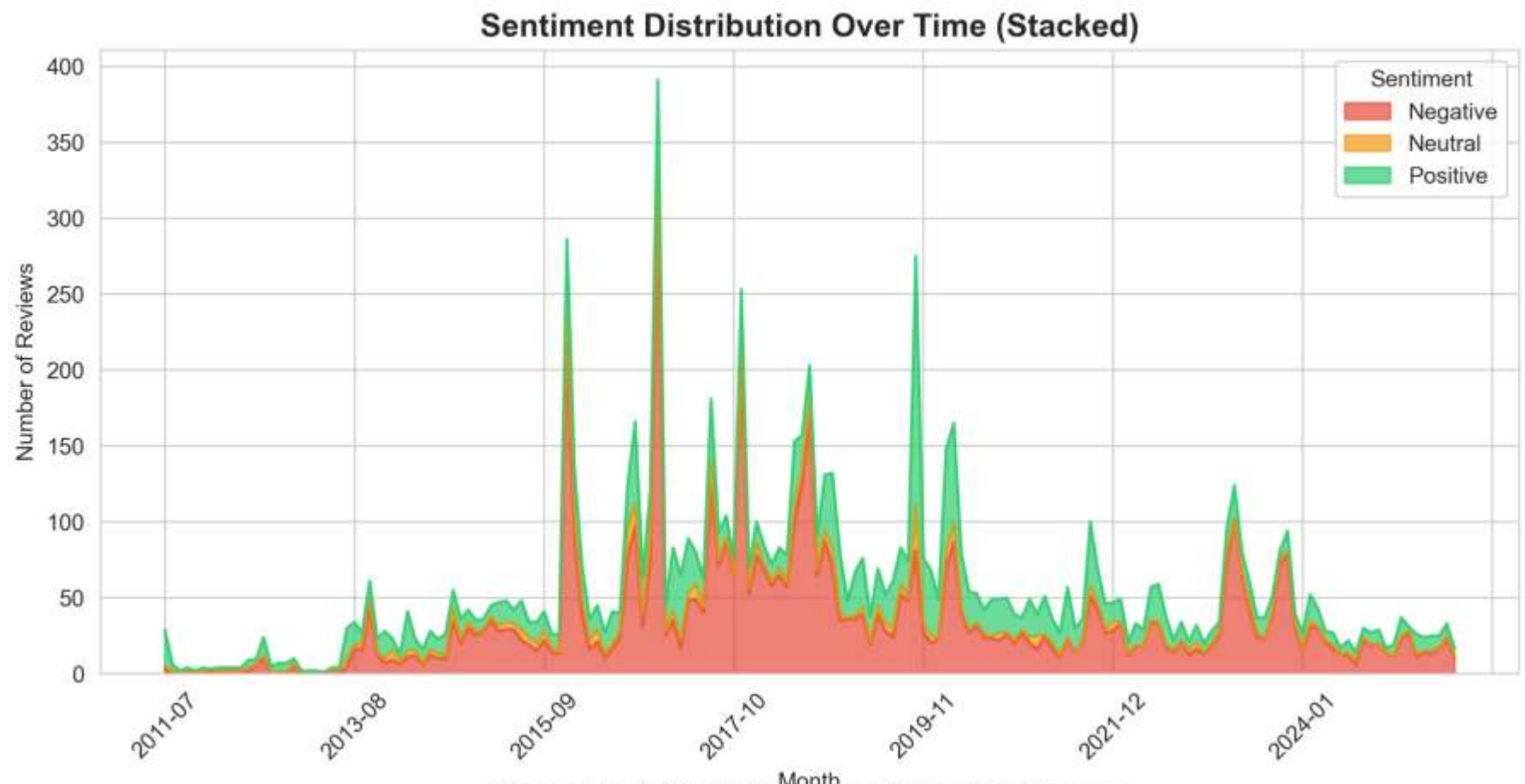


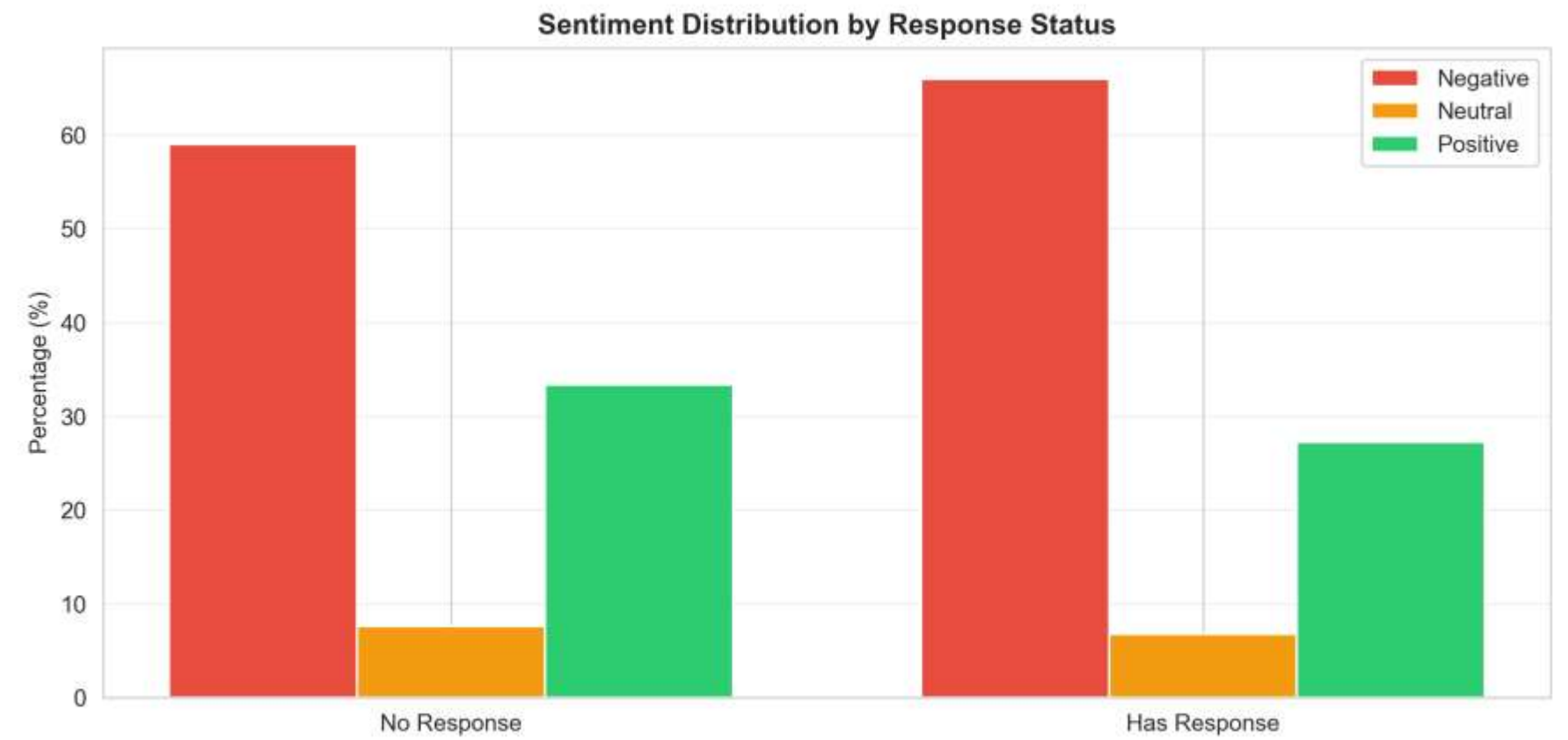
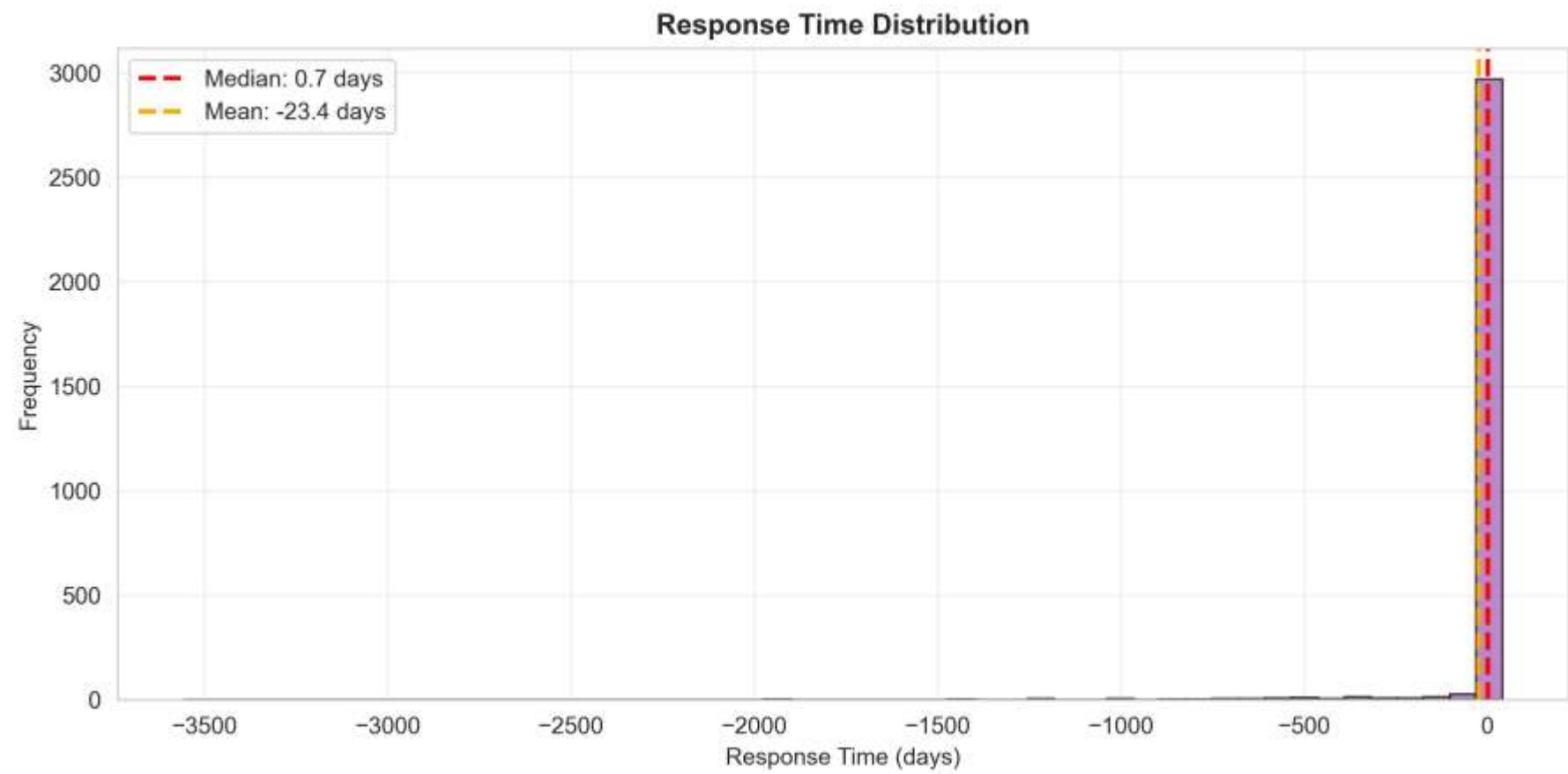
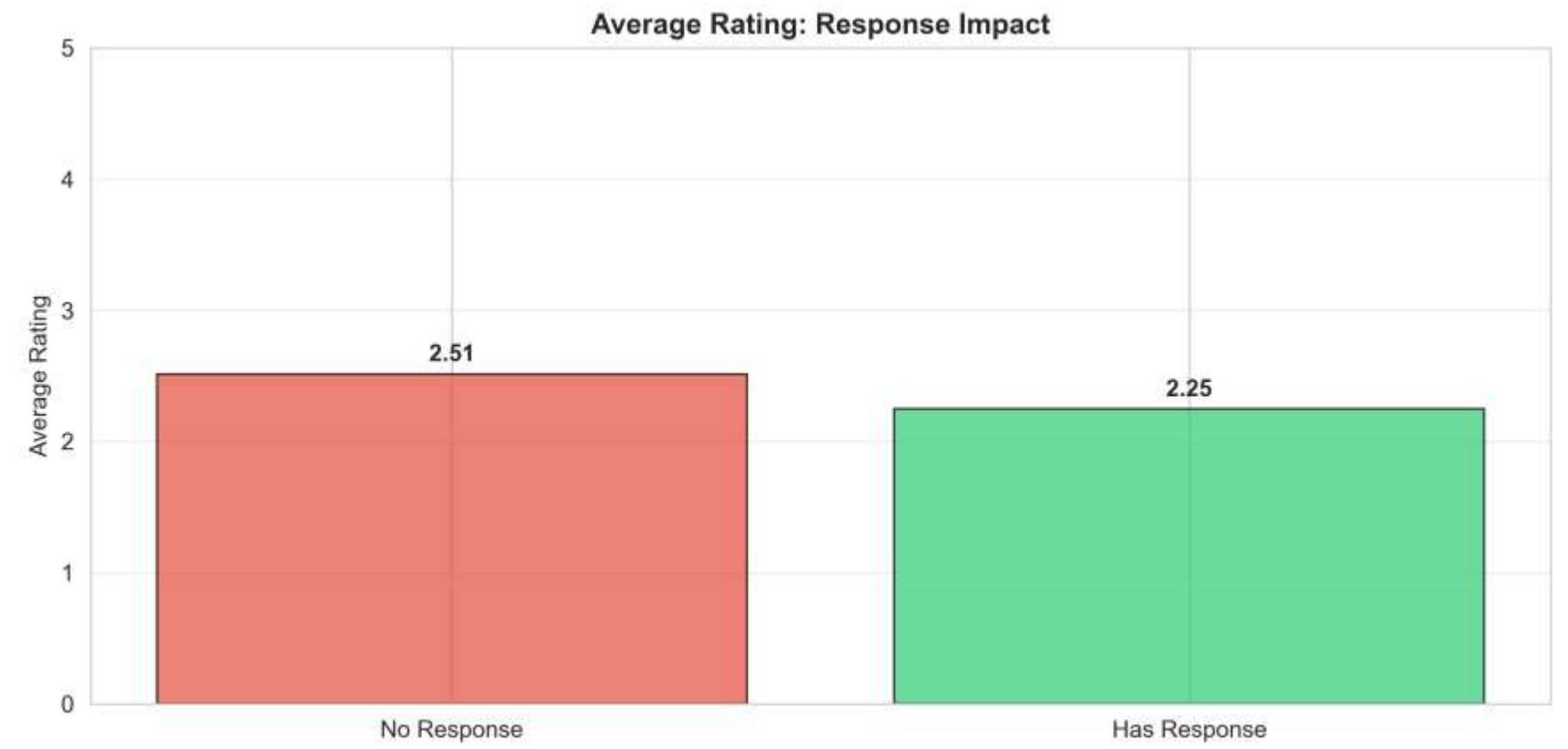
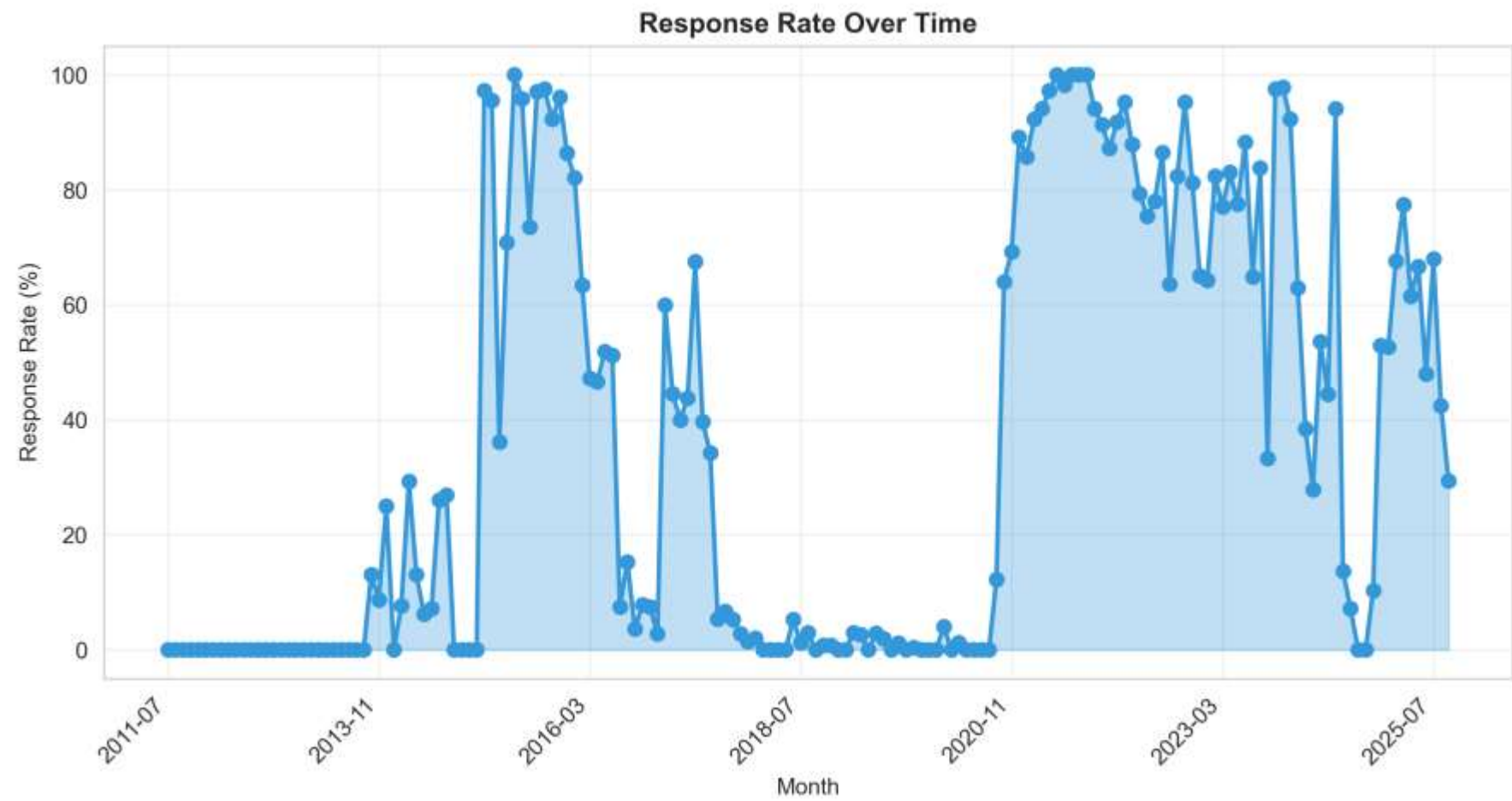
Response Rate Impact



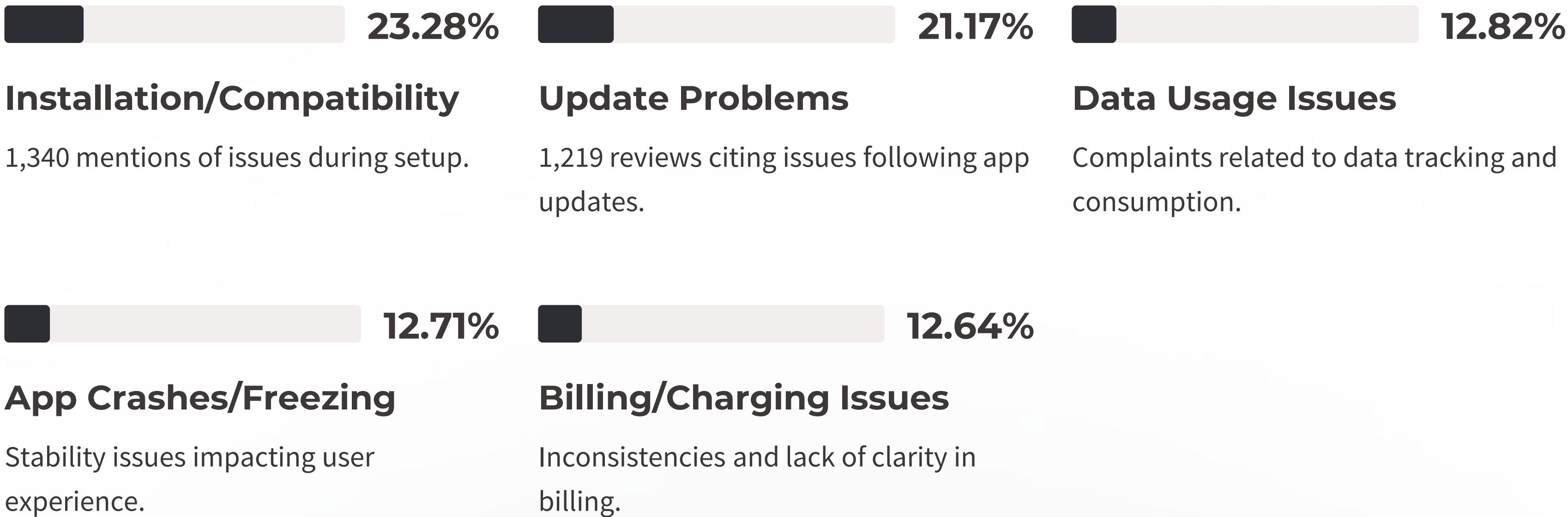
Sentiment by App Version







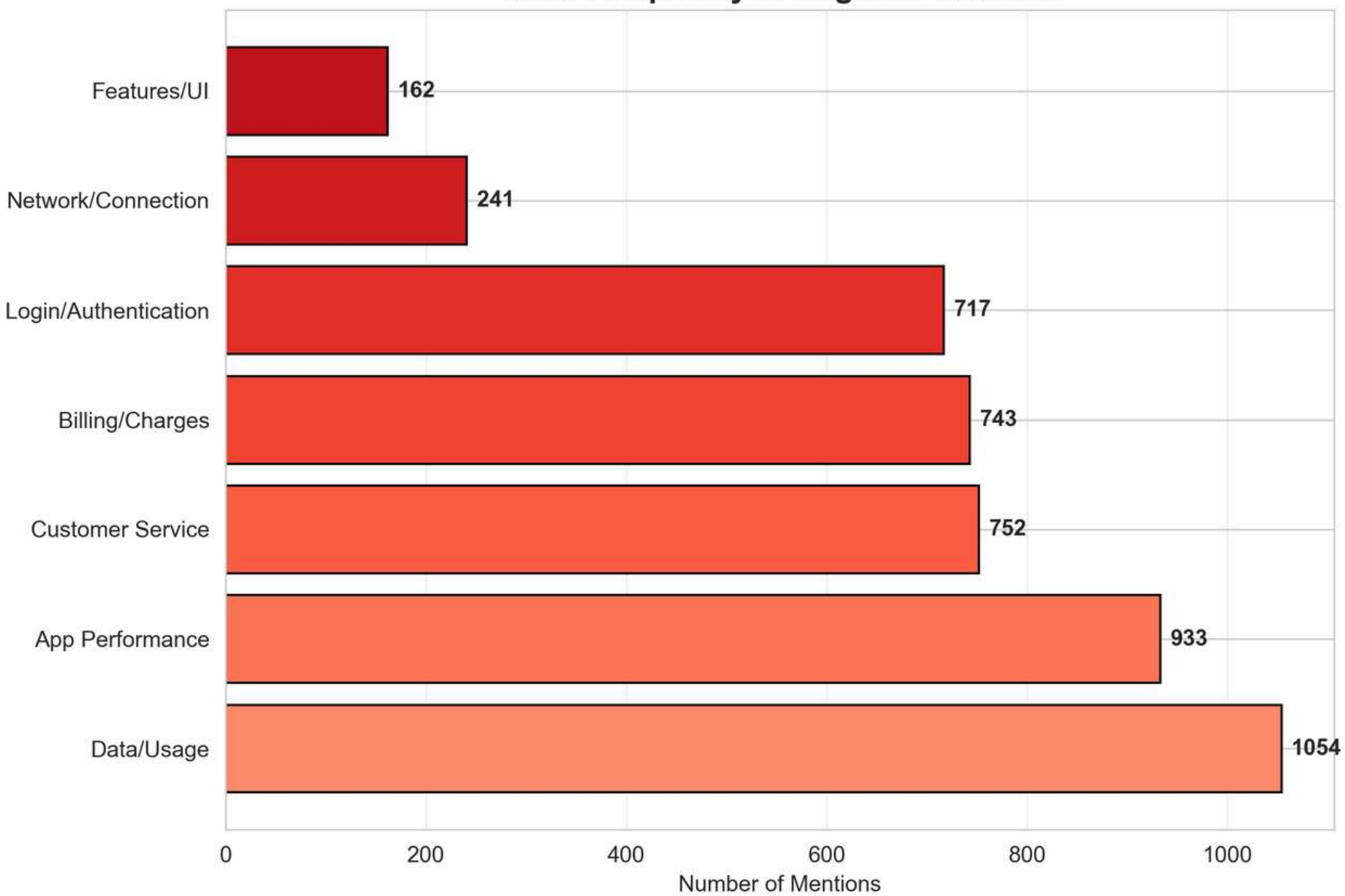
Top Issues in Negative Feedback



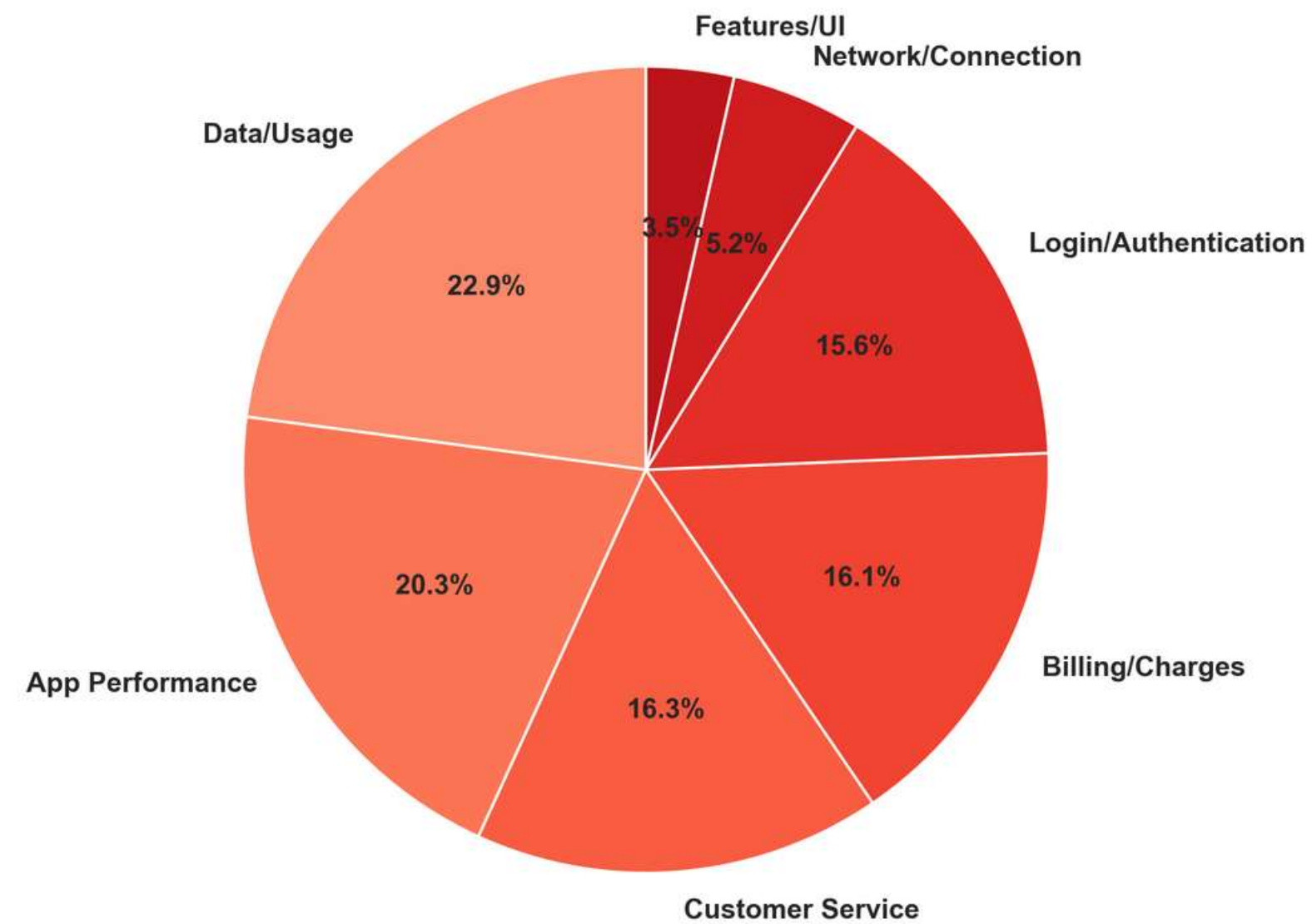
Detailed Issue Breakdown:

Issue Category	Mentions	Percentage
Installation/Compatibility	1340	23.280056
Update Problems	1219	21.177901
Data Usage Issues	738	12.821404
App Crashes/Freezing	732	12.717165
Billing/Charging Issues	728	12.647672
Customer Service Issues	680	11.813760
Account Issues	499	8.669215
Feature Not Working	475	8.252259
Balance/Credit Issues	438	7.609451
User Interface Problems	339	5.889507
Network/Connection Issues	316	5.489924
Login/Authentication Issues	311	5.403058
Slow Performance	254	4.412787
Missing Features	111	1.928423

Issue Frequency in Negative Reviews



Issue Distribution (% of Negative Reviews)



Topic Modeling

Topic modeling identified five dominant themes.

Topic 0:

good, easy, use, easy use, download, happy, thank, fine, uninstall, simple

Topic 1:

update, time, working, data, like, deal, work, load, open, need

Topic 2:

awesome, thanks, service, bad, slow, helpful, customer, best, apps, want

Topic 3:

work, doesnt, useless, doesnt work, nice, open, bill, pay, never, cool

Topic 4:

great, update, love, log, time, data, keep, much, say, let

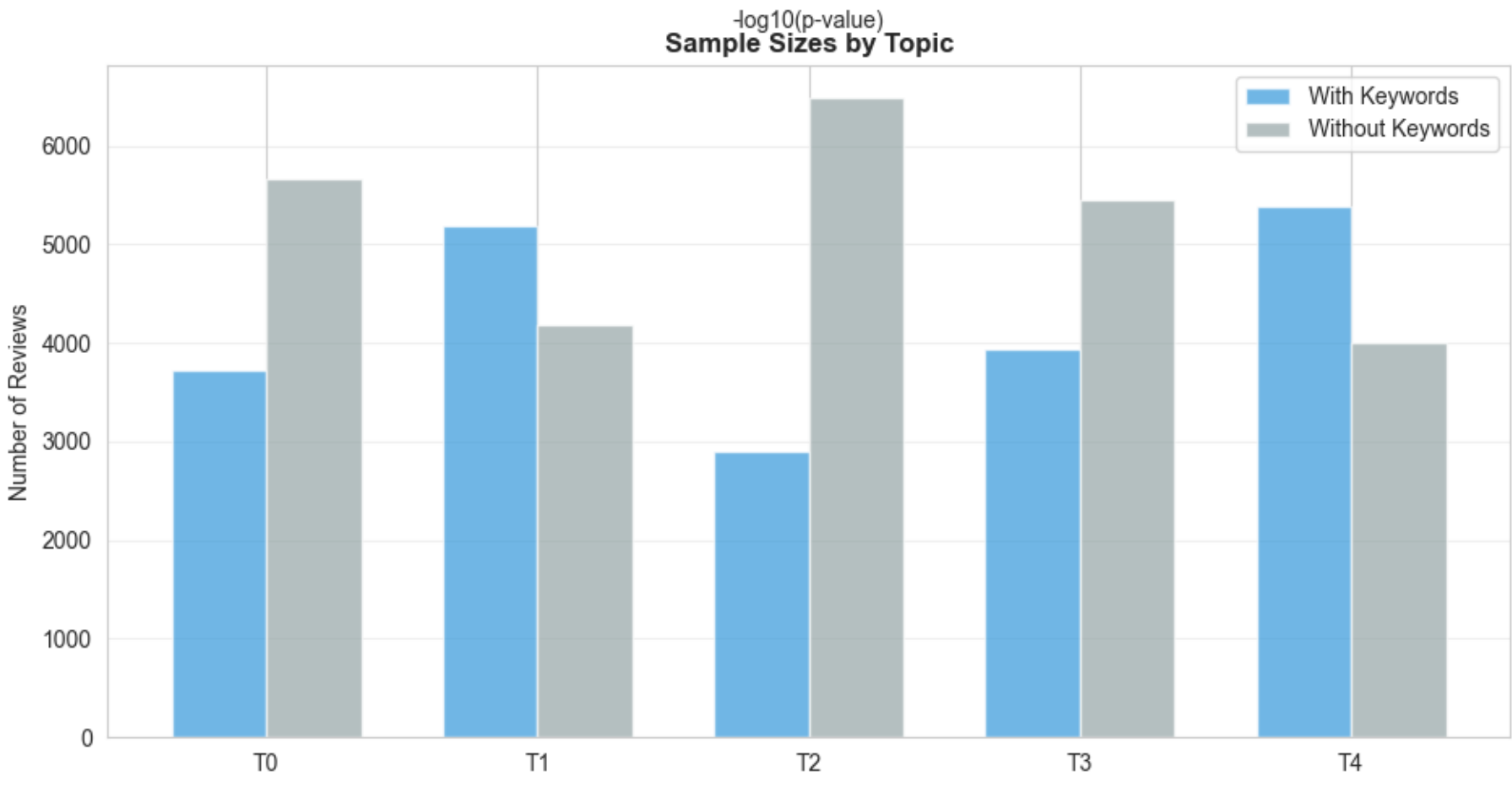
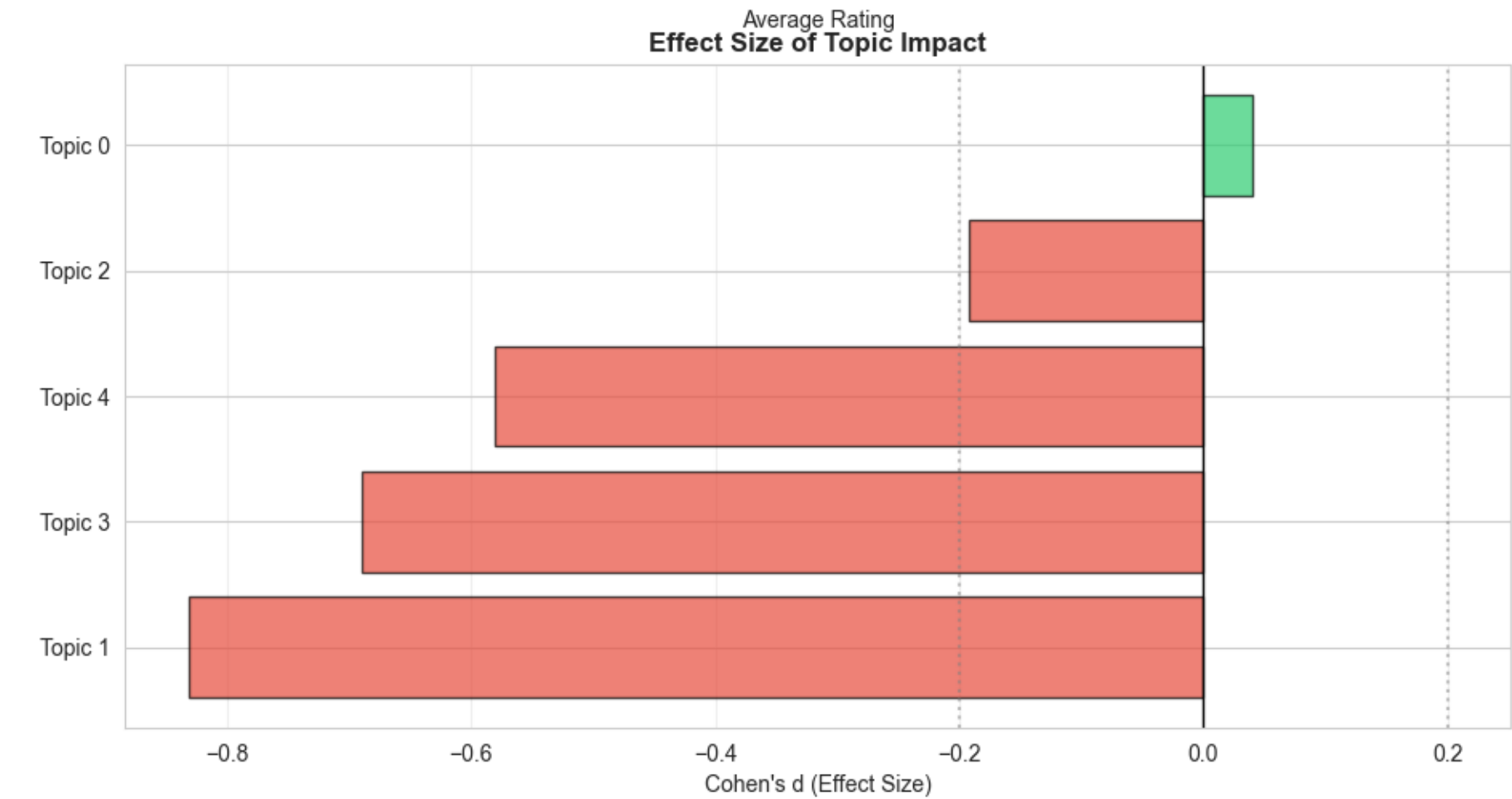
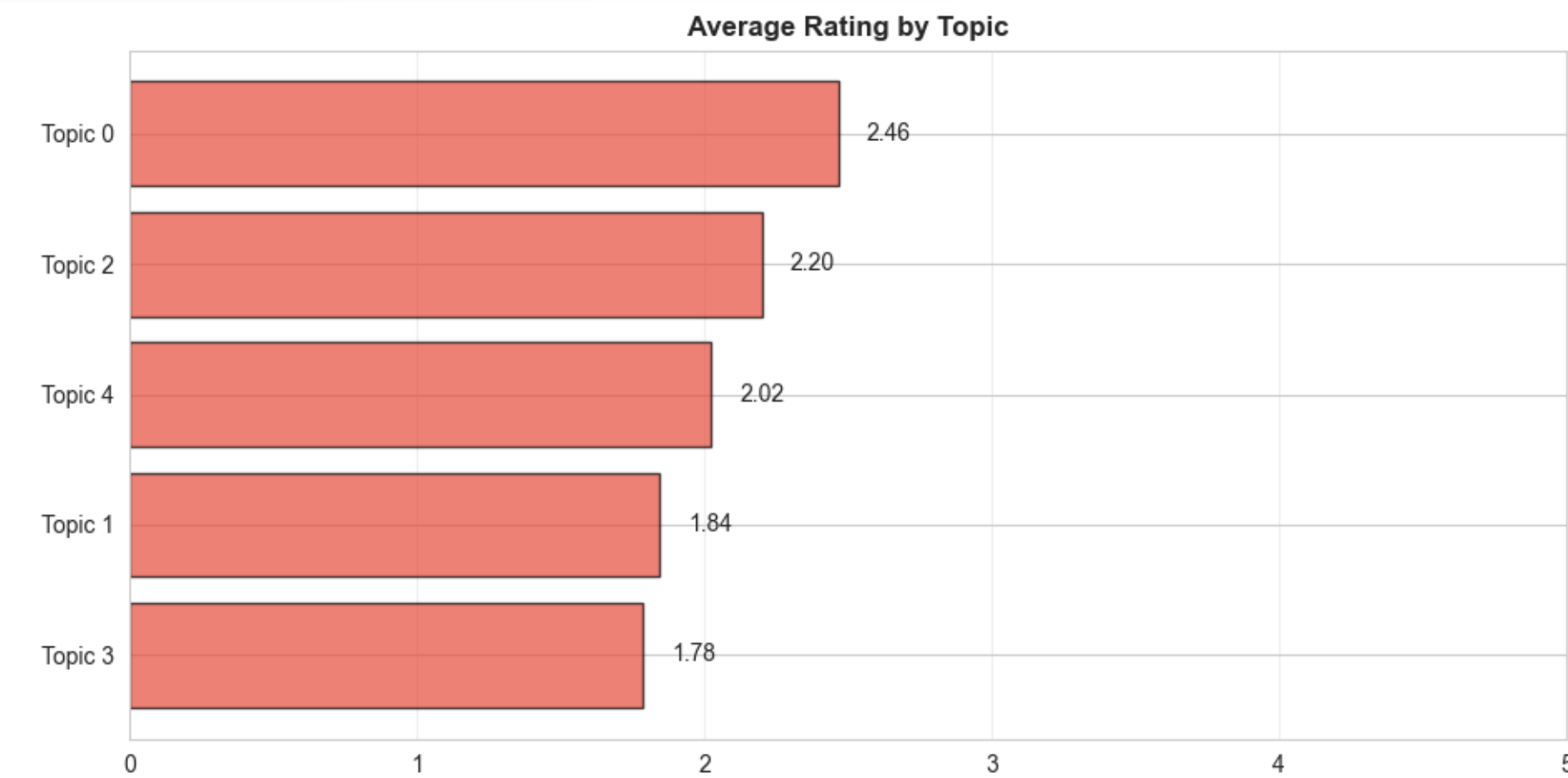
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Statistical Significance

- **Topic 0** shows no significant correlation ($d = 0.04$, $p = 0.059$)
- **Topic 1 (Updates/Working)** shows the strongest negative impact with 5,195 reviews (55.3%) average 1.84/5.0 versus 3.14/5.0 for others
- **Topic 2 (Service Quality)** shows negligible practical impact ($d = -0.19$, - 0.32 stars).
- **Topic 3 (Non-Functionality)** affects 3,939 reviews (41.9%), averaging 1.78/5.0 versus 2.89/5.0 for others, a 1.10-star decrease ($d = -0.69$, medium-large, $p = 0$).
- **Topic 4 (Sarcasm)** consisting of 5,383 reviews (57.3%) and has an average of 2.02/5.0 despite positive words, a decrease of 0.94-stars ($d = -0.58$, medium, $p = 0$)

These incredibly small p-values ($1.63e-302$, $1.84e-246$, $4.82e-155$) attest beyond any doubt that these differences are true differences and not due to random chance

The three major issues appear interconnected, recent application updates introduced bugs therefore breaking functionality and prompting users to use sarcastic language to express frustration



Version Analysis: The Catastrophic Rating Decline

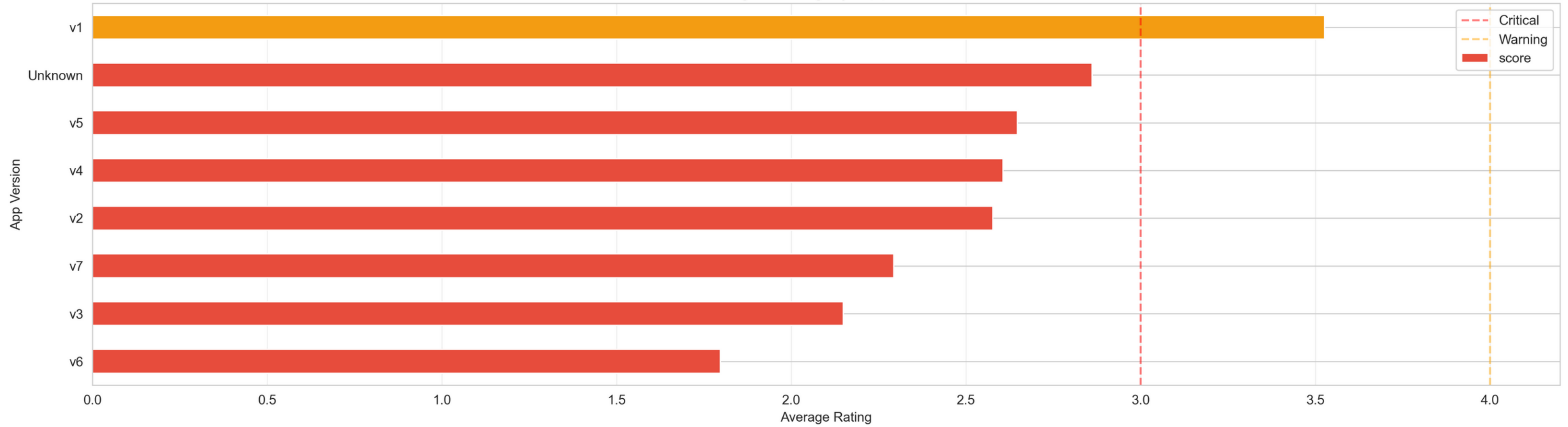
Version 7.x

Version-level analysis shows a dramatic decline in user satisfaction, suggesting systematic defects introduced in recent updates.

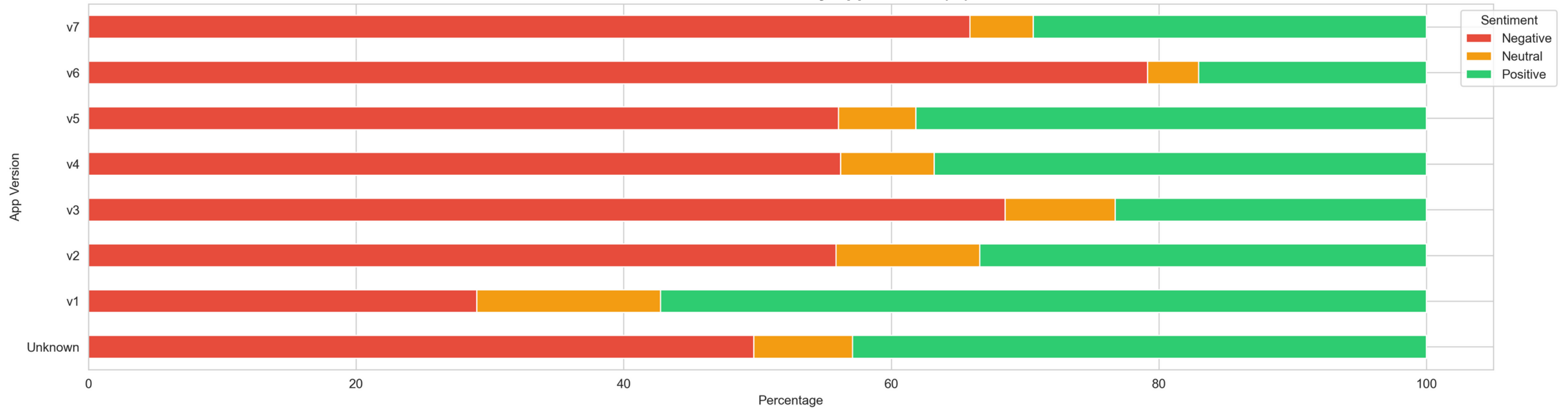
The top priority is to stabilize the update and release process to prevent further degradation.

- **Version 7.5 Average Rating:** 2.45 (64.7% Negative)
- **Baseline Version 1.x Rating:** 3.78 (35.1% Positive)
- **Version 3.x:** Most negative reviews

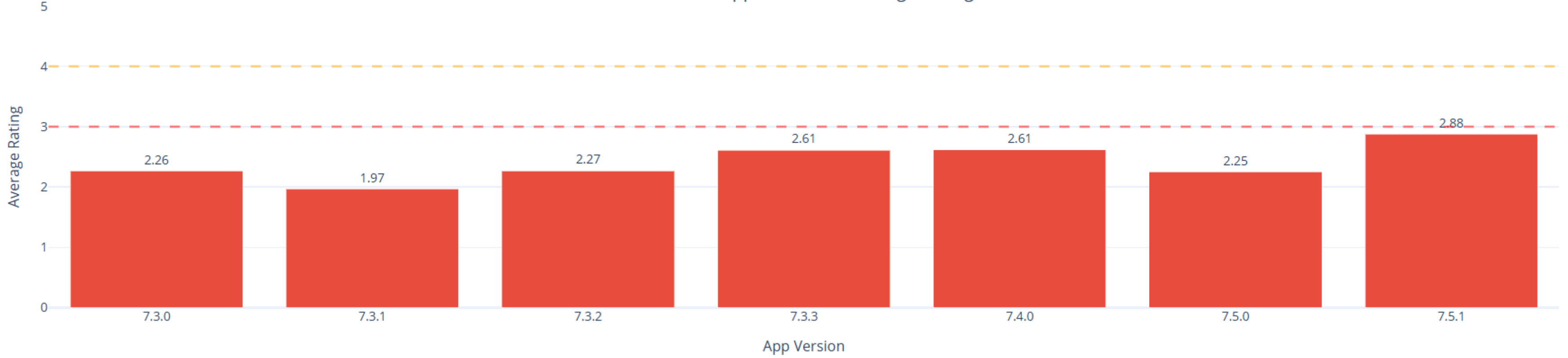
Average Rating by App Version



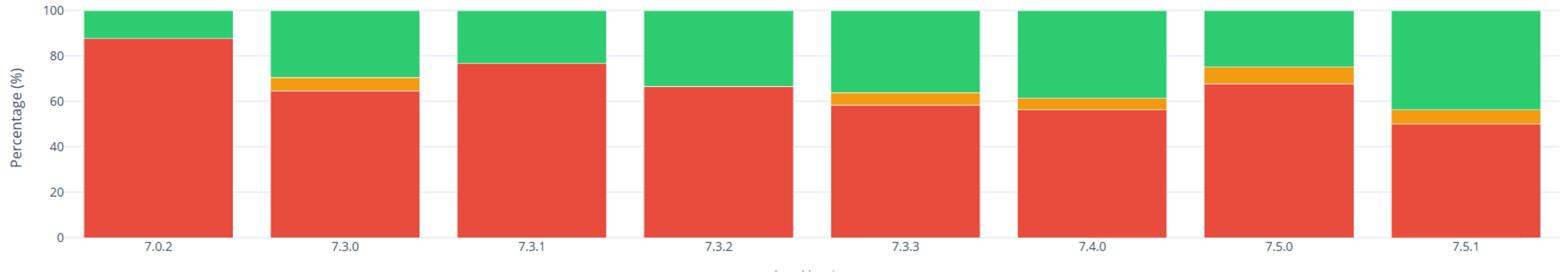
Sentiment Distribution by App Version (%)

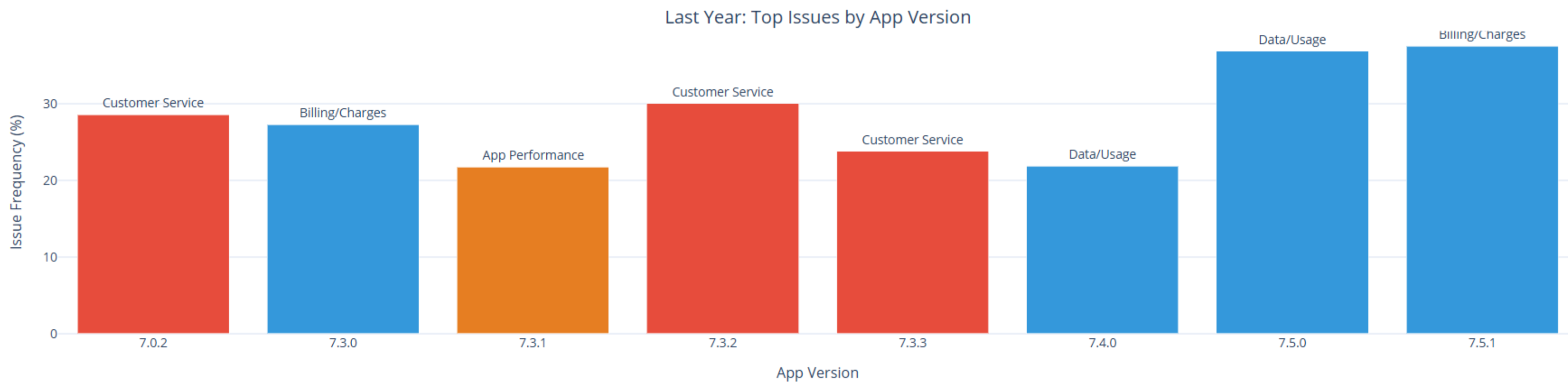


Last Year: App Version vs Average Rating



Last Year: Review Distribution by App Version

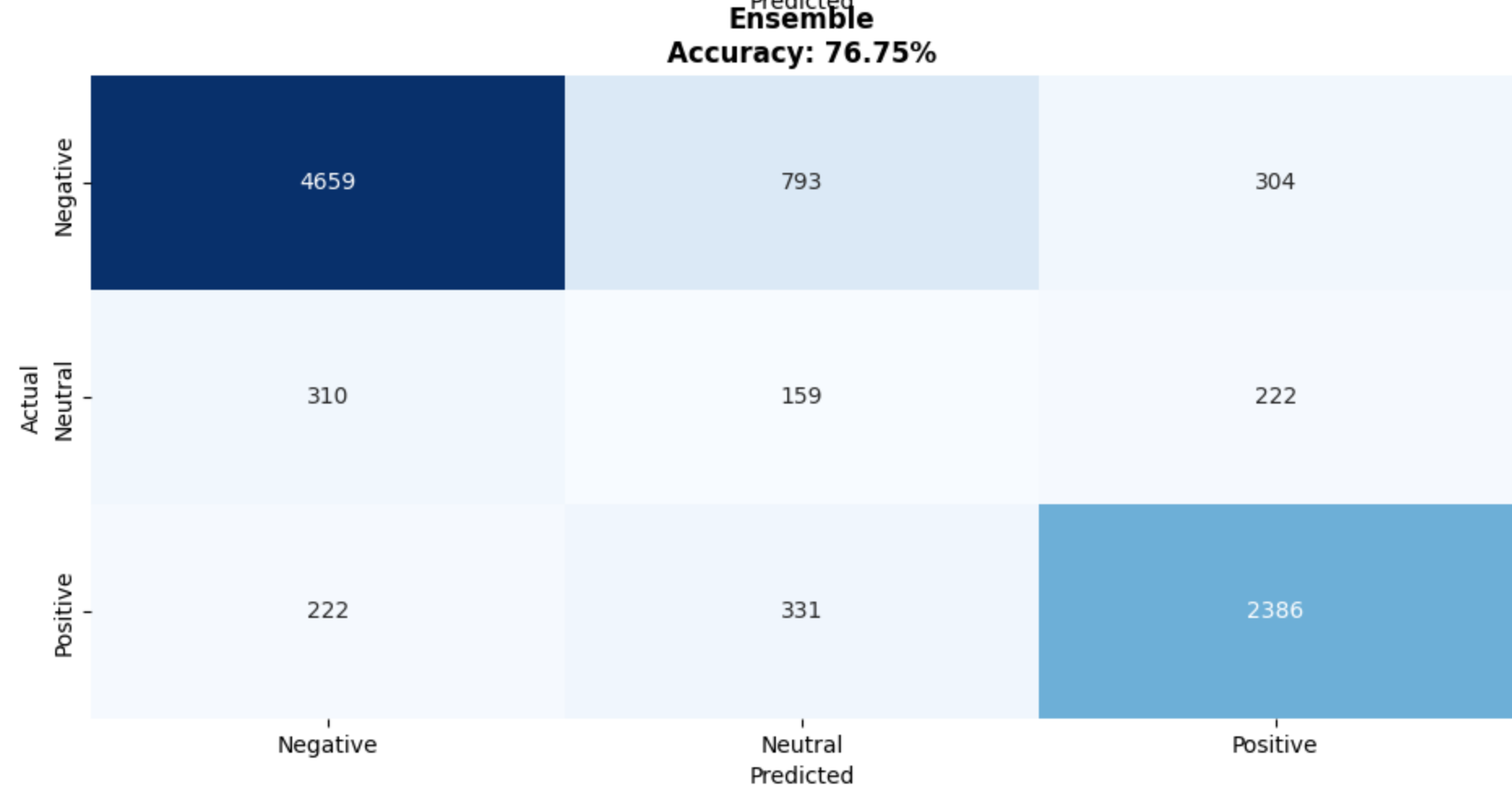
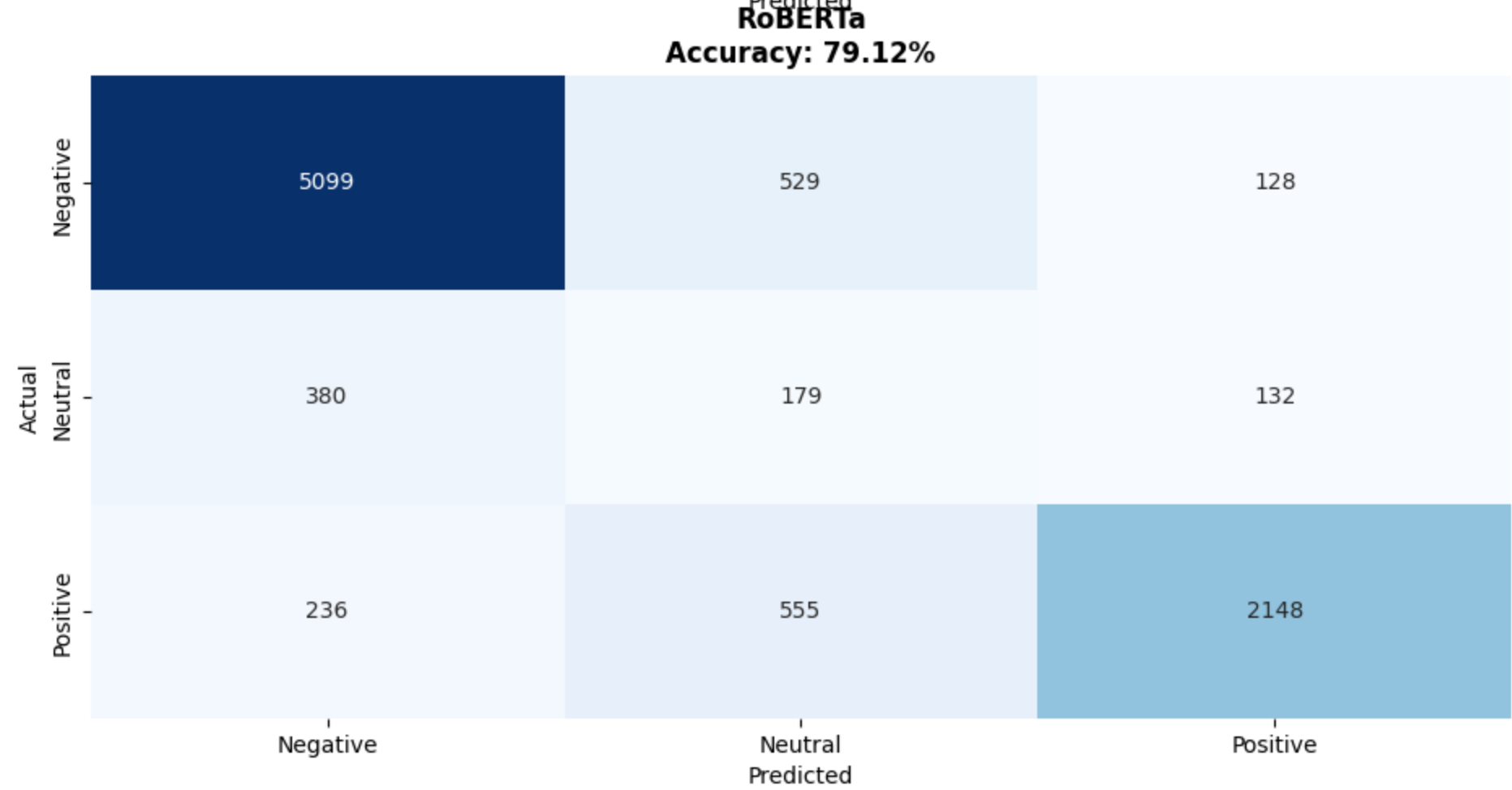
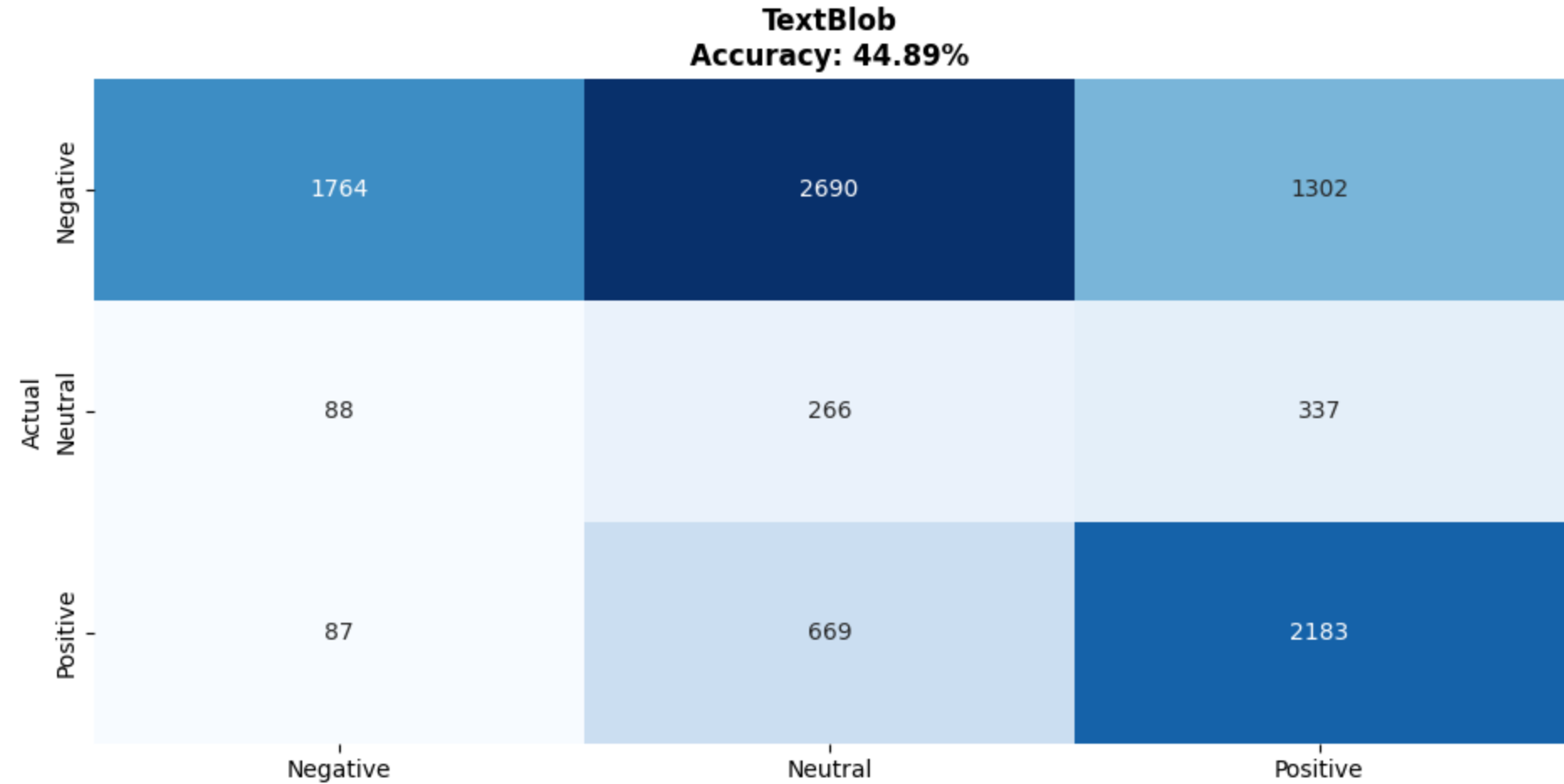
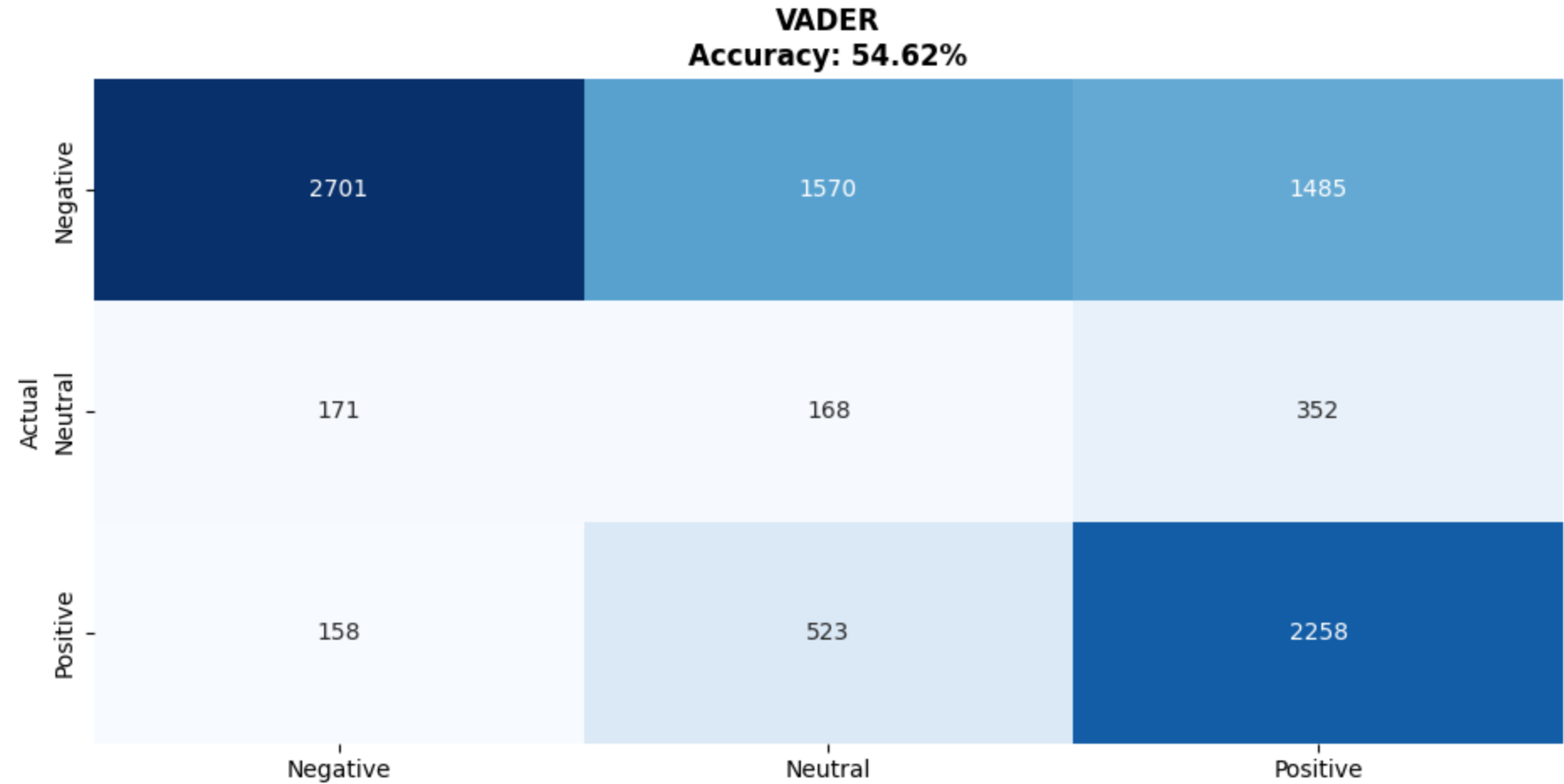




Unsupervised Sentiment Classification

RoBERTa reached 79.1% accuracy, improvement from VADER, and fairly consistent precision in all the classes, namely, negative 88.58%, neutral 25.90% and positive 72.83%.

Much lower precision of neutral class (25.90%) also corresponds to an innate difficulty of neutral sentiment classification, which itself incorporates true mixed feelings and mild positive/negative sentiments



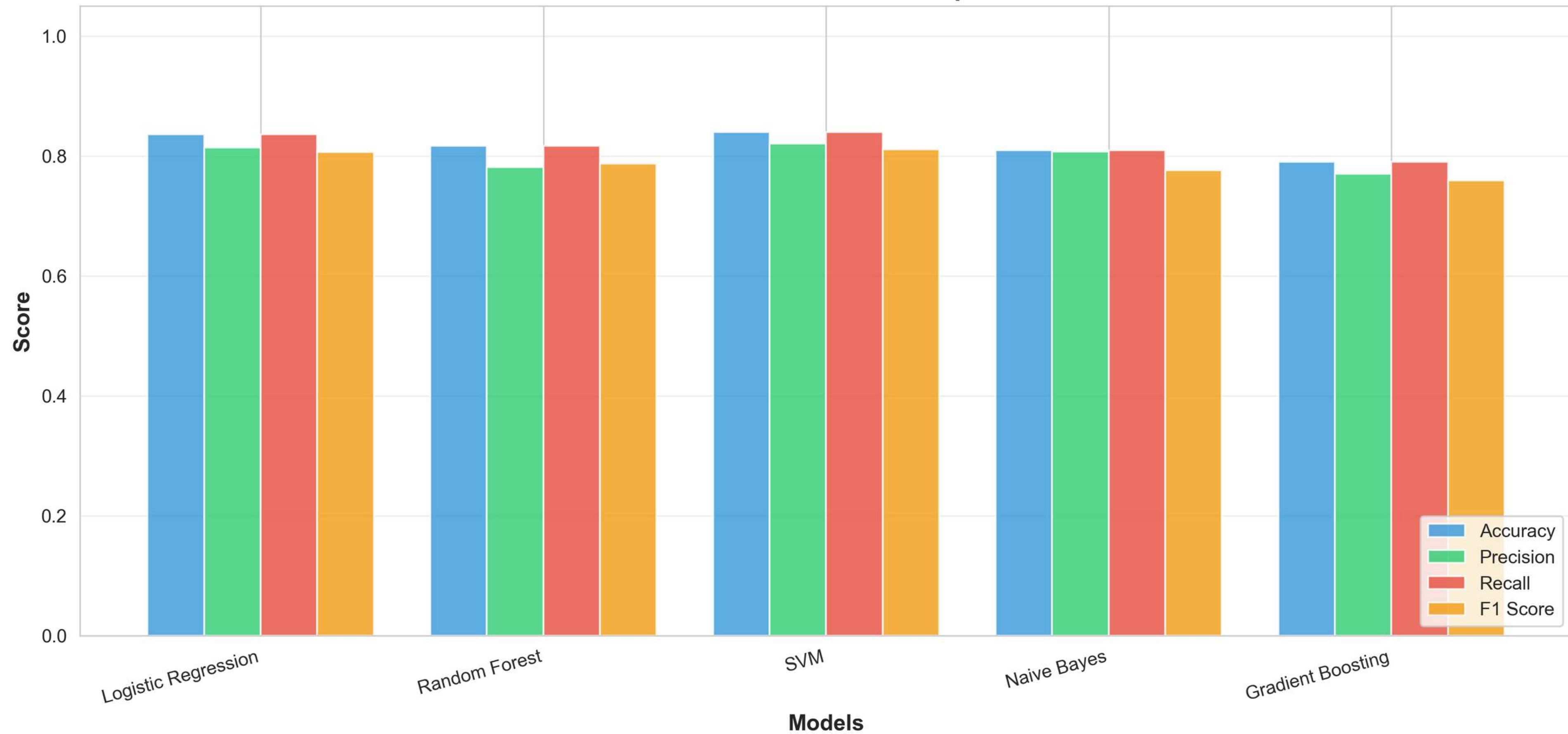
Model Performance

The Support Vector Machine (SVM) model achieved the highest accuracy, demonstrating strong potential for automated sentiment monitoring.

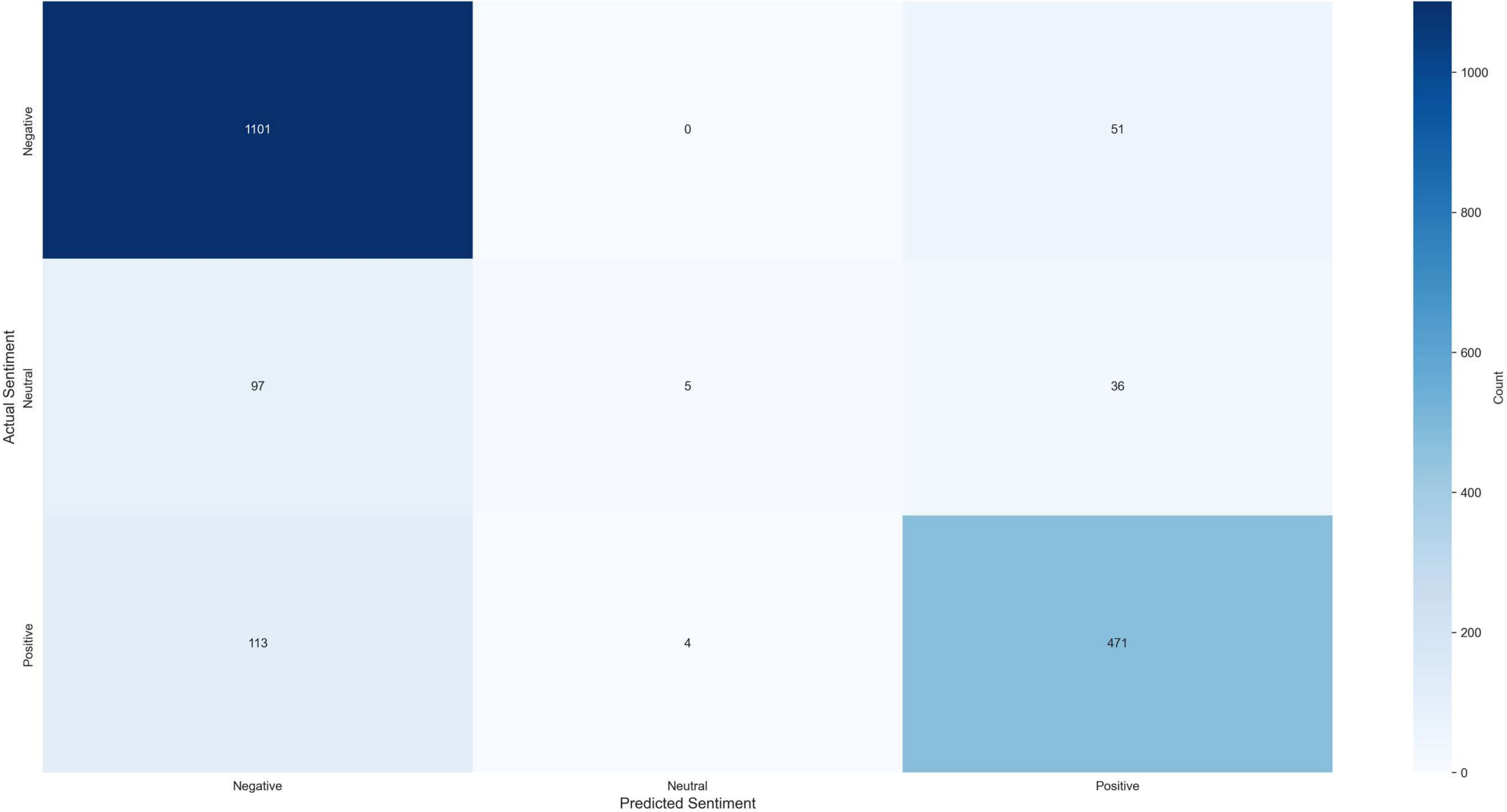
Model	Accuracy	Precision	Recall	F1-Score	Training Time
SVM	0.8397	0.8203	0.8397	0.8108	20.91 s
Logistic Regression	0.8360	0.8136	0.8360	0.8061	0.24 s
Random Forest	0.8168	0.7812	0.8168	0.7867	7.76 s
Naive Bayes	0.8094	0.8068	0.8094	0.7757	0.01 s
Gradient Boosting	0.7902	0.7697	0.7697	0.7586	18.49 s

Table 1: Machine Learning Model Performance Comparison

ML Model Performance Comparison



Confusion Matrix - SVM





7. Ethical Practices and Limitations

Data Privacy and Ethics

Analysis used only publicly available reviews, excluding PII. Anonymization was applied to usernames, and methodological transparency was maintained for reproducibility.

Algorithmic Bias Mitigation

Bias was mitigated using ensemble techniques and validation against ground truth. Interpretable models (SVM) were favored for transparency.

Limitations

Single-platform focus (Google Play only) and self-selection bias in reviewers limit generalizability. Correlation does not imply causation.

Measurement Error

Sentiment accuracy of 79.12% means approximately 1 in 6 reviews is misclassified, contributing to measurement error.

6. Strategic Recommendations



Stabilize Update Process

Implement staged rollouts and increased testing. This could address ~70% of complaints and raise ratings from 2.42 to 3.72/5.0.



Response Rate Recovery

Staff managers and train staff for empathy and technical escalation. Set a response time goal of <24 hours (down from 7-day median).



Billing Transparency

Enhance in-app billing with itemized charges, proactive notifications (3-day advance notice), and simplified dispute resolution.



Beta Testing Program

Shift from reactive firefighting to proactive validation. Incentivize negative reviewers to test and provide feedback.



Long-Term Strategy: Modernization & Quality Culture

Infrastructure Modernization

Address technical debt for a sustainable platform.

Implement CDN for geographic optimization.

Ensure Infrastructure Redundancy (99.99% uptime target).

Adopt Microservices Architecture for independent updates.

Quality-First Cultural Transformation

Establish a specialized QA Team.

Define a clear "Definition of Done."

Implement Executive Quality Reviews.

Predictive Analytics Platform

Automated Sentiment Monitoring for real-time issue detection.

Predictive Churn Modeling to identify at-risk customers.

Feature Request Extraction to prioritize enhancements.

Competitive Intelligence tracking.

Innovation Lab

Focus on competitive differentiation.

Explore Augmented Reality for service visualization.

Develop Voice Control for hands-free operation.

Ethical Considerations & Limitations

Ethical Data Handling

Adhered to strict data privacy: using only public, anonymized data, reporting aggregated statistics, and avoiding individual user identification or demographic inference.

Algorithmic Bias Mitigation

Mitigation strategies address Demographic Bias (ensemble methods), Language Bias (social media-trained RoBERTa), and Domain Adaptation (validation against in-domain ground truth).

Study Limitations

Platform Coverage

Only Google Play reviews were analyzed; iOS user differences are unknown.

Representativeness

Self-selection bias means reviewers are a motivated subset; the "silent majority" is not represented.

Causation vs. Correlation

Observational design cannot prove causation; confounding factors may exist.

Classification Accuracy

79.12% accuracy means measurement error is present (~1 in 6 reviews misclassified).

THE END

